

**ULTIMATE SOLUTION BOOK:
ALL EXERCISES & PAST EXAMS**

Introduction to Methods in Operational Research

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

2027 EDITION

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About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- ✓ My name is **Jovan Samardzic**, I am 26 years old
- ✓ Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- ✓ Recipient of the **Science@Leuven Scholarship**

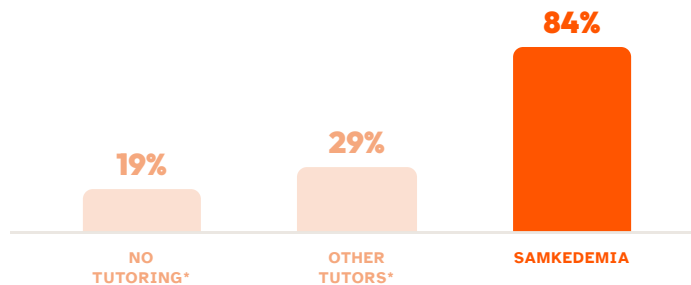
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Chapter 1 - Indefinite Integrals

Question 1

Show that F is an anti-derivative of f .

(a) $F(x) = \frac{x^2}{x+1}, \quad f(x) = \frac{x^2 + 2x}{(x+1)^2}$

(b) $F(x) = x - \ln(1 + e^x), \quad f(x) = \frac{1}{e^x + 1}$

Tip: Verification via Differentiation

To show that $F(x)$ is an anti-derivative of $f(x)$, follow these steps:

1. **Differentiate $F(x)$:** Compute $F'(x)$ using the appropriate rules (Quotient Rule, Chain Rule, etc.).
2. **Simplify:** Algebraically manipulate $F'(x)$ to match the form of $f(x)$.
3. **Conclude:** If $F'(x) = f(x)$, then $F(x)$ is an anti-derivative of $f(x)$.

✓ **Solution 1: (a)** $F(x) = \frac{x^2}{x+1}, \quad f(x) = \frac{x^2 + 2x}{(x+1)^2}$

The function f is defined for all $x \neq -1$. We apply the Quotient Rule:

$$\begin{aligned} F'(x) &= \frac{(2x)(x+1) - (x^2)(1)}{(x+1)^2} \\ &= \frac{2x^2 + 2x - x^2}{(x+1)^2} \\ &= \frac{x^2 + 2x}{(x+1)^2} \end{aligned}$$

Since $F'(x) = f(x)$, F is an anti-derivative of f .

Answer: $F'(x) = \frac{x^2+2x}{(x+1)^2}$

✓ **Solution 1: (b)** $F(x) = x - \ln(1 + e^x), \quad f(x) = \frac{1}{e^x + 1}$

Differentiating $F(x)$ term by term:

$$\begin{aligned} F'(x) &= (x)' - (\ln(1 + e^x))' \\ &= 1 - \frac{1}{1 + e^x} \cdot (1 + e^x)' \\ &= 1 - \frac{1}{1 + e^x} \cdot e^x \\ &= \frac{1 + e^x}{1 + e^x} - \frac{e^x}{1 + e^x} \end{aligned}$$

Simplifying the expression:

$$F'(x) = \frac{1 + e^x - e^x}{1 + e^x} = \frac{1}{1 + e^x}$$

Since $F'(x) = f(x)$, F is an anti-derivative of f .

Answer: $F'(x) = \frac{1}{e^x+1}$

Question 2

Show that:

$$(a) \int \frac{x^2 dx}{\sqrt{1+x}} = \frac{2}{15}(3x^2 - 4x + 8)\sqrt{1+x} + C$$

$$(b) \int \frac{du}{2^u + 3} = \frac{u}{3} - \frac{1}{3 \ln 2} \ln(2^u + 3) + C$$

✓ Solution 2: (a) $\int \frac{x^2 dx}{\sqrt{1+x}} = \frac{2}{15}(3x^2 - 4x + 8)\sqrt{1+x} + C$

Let $F(x) = \frac{2}{15}(3x^2 - 4x + 8)(1+x)^{1/2}$. We apply the Product Rule:

$$\begin{aligned} F'(x) &= \frac{2}{15} \left[(6x - 4)\sqrt{1+x} + (3x^2 - 4x + 8) \frac{1}{2\sqrt{1+x}} \right] \\ &= \frac{2}{15} \left[\frac{2(6x - 4)(1+x) + (3x^2 - 4x + 8)}{2\sqrt{1+x}} \right] \\ &= \frac{1}{15\sqrt{1+x}} \left[2(6x^2 + 2x - 4) + 3x^2 - 4x + 8 \right] \\ &= \frac{1}{15\sqrt{1+x}} \left[12x^2 + 4x - 8 + 3x^2 - 4x + 8 \right] \\ &= \frac{15x^2}{15\sqrt{1+x}} = \frac{x^2}{\sqrt{1+x}} \end{aligned}$$

Since $F'(x)$ matches the integrand, the equality is shown.

Answer: $F'(x) = \frac{x^2}{\sqrt{1+x}}$

✓ **Solution 2: (b)** $\int \frac{du}{2^u + 3} = \frac{u}{3} - \frac{1}{3 \ln 2} \ln(2^u + 3) + C$

Let $F(u) = \frac{u}{3} - \frac{1}{3 \ln 2} \ln(2^u + 3)$. Differentiating with respect to u :

$$\begin{aligned} F'(u) &= \frac{1}{3} - \frac{1}{3 \ln 2} \cdot \frac{1}{2^u + 3} \cdot (2^u \ln 2) \\ &= \frac{1}{3} - \frac{2^u \ln 2}{3 \ln 2 (2^u + 3)} \\ &= \frac{1}{3} - \frac{2^u}{3(2^u + 3)} \end{aligned}$$

Finding a common denominator:

$$F'(u) = \frac{(2^u + 3) - 2^u}{3(2^u + 3)} = \frac{3}{3(2^u + 3)} = \frac{1}{2^u + 3}$$

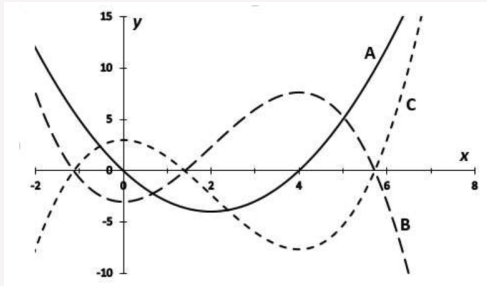
Since $F'(u)$ matches the integrand, the equality is shown.

Answer: $F'(u) = \frac{1}{2^u + 3}$

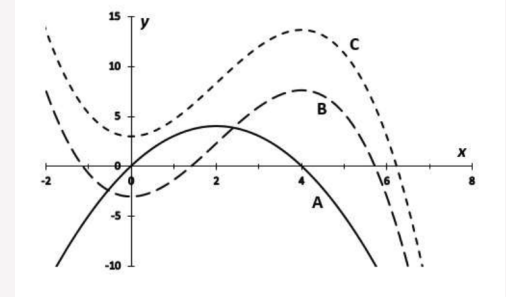
Question 3

In each of the following figures, three graphs are shown. If A is the graph of the function f , which of the other two graphs is the graph of an anti-derivative of f ?

a.



b.



Tip: Relationship Between a Function and its Anti-derivative

To identify the graph of an anti-derivative F from the graph of f , follow these steps:

1. **Check the Zeroes:** When the graph of f is at zero (crosses the x -axis), the graph of F must have a local maximum or minimum (a flat turning point).
2. **Check the Sign:** If f is positive (above the x -axis), F must be increasing. If f is negative (below the x -axis), F must be decreasing.
3. **Check the Peaks:** When f has a peak or valley, F changes its curvature (inflection point).

✓ Solution 3: (a) Identifying the anti-derivative for graph A

In figure (a), graph A represents the function f . We look at where graph A crosses the x -axis:

- Graph A is zero at $x \approx 0.5$ and $x = 4$.
- At $x \approx 0.5$, graph C has a local minimum (it turns upward).
- At $x = 4$, graph C has a local maximum (it turns downward).

- Graph B does not have turning points at these specific x -values.

Additionally, between $x \approx 0.5$ and $x = 4$, graph A is positive, and graph C is increasing during this entire interval.

Answer: Graph C

✓ Solution 3: (b) Identifying the anti-derivative for graph A

In figure (b), graph A is a downward-opening parabola representing f .

- Graph A crosses the x -axis at $x = 0$ and $x = 4$.
- Both graph B and graph C have turning points at exactly $x = 0$ (local minimums) and $x = 4$ (local maximums).
- Between $x = 0$ and $x = 4$, graph A is positive, and both B and C are increasing.

Because B and C have the exact same shape and only differ by a vertical shift (a constant C), they both represent valid anti-derivatives of the same function f .

Answer: Both B and C

? Question 4

Why is the formula $\int r f(x) dx = r \int f(x) dx$ not correct when $r = 0$?

💡 Tip: Constants in Integration

When working with indefinite integrals and constants, keep these rules in mind:

1. **Integral of Zero:** The integral of zero is an arbitrary constant C , because the derivative of any constant is zero.
2. **Multiplication by Zero:** Multiplying any finished expression by zero results in exactly zero, which removes the constant of integration.

✓ Solution 4

We compare the two sides of the equation when the constant r is zero:

- **Left-Hand Side:** The multiplication happens *inside* the integral before integration:

$$\int 0 \cdot f(x) dx = \int 0 dx = C$$

The result is an arbitrary constant C , which represents any real number.

- **Right-Hand Side:** The multiplication happens *after* the integration is performed:

$$0 \cdot \int f(x) dx = 0 \cdot (F(x) + C) = 0$$

The result is exactly zero.

Since an arbitrary constant C is not necessarily equal to zero, the two sides are not equal. The formula fails in this specific case because it "loses" the constant of integration on the right side.

Answer: $C \neq 0$

Question 5

Calculate using integrals of basic functions and basic integration rules:

a. $\int \frac{1}{x^4} dx$

b. $\int \sqrt[3]{x} dx$

c. $\int (t+1)\sqrt{t} dt$

d. $\int (2^x + 3^x) dx$

e. $\int \frac{2^x}{3^x} dx$

f. $\int \frac{x^3 + x - 1}{\sqrt[3]{x}} dx$

g. $\int (\sqrt{x} + 1)(x - \sqrt{x}) dx$

h. $\int \left(x^3 - \frac{7}{x} + e^x \right) dx$

i. $\int \frac{\sqrt{t} - t + t^3}{\sqrt[3]{t}} dt$

j. $\int (x^e + e^x) dx$

k. $\int e^{\sqrt{2}} dx$

Tip: Basic Integration Rules

To solve these problems, we apply the basic building blocks of integration.

Power Rule	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$	$n \neq -1$
Logarithmic	$\int \frac{1}{x} dx = \ln x + C$	
General Exponential	$\int a^x dx = \frac{a^x}{\ln a} + C$	

✓ **Solution 5: (a)** $\int \frac{1}{x^4} dx$

Rewrite using a negative exponent:

$$\int x^{-4} dx$$

Apply the power rule $\int x^n dx = \frac{x^{n+1}}{n+1} + C$:

$$\frac{x^{-4+1}}{-4+1} + C = \frac{x^{-3}}{-3} + C = -\frac{1}{3x^3} + C$$

Answer: $-\frac{1}{3x^3} + C$

✓ **Solution 5: (b)** $\int \sqrt[3]{x} dx$

Rewrite the radical as a fractional power:

$$\int x^{1/3} dx$$

Apply the power rule:

$$\frac{x^{1/3+1}}{1/3+1} + C = \frac{x^{4/3}}{4/3} + C = \frac{3}{4}x^{4/3} + C$$

Convert back to radical form:

$$\frac{3}{4}\sqrt[3]{x^4} + C = \frac{3}{4}x\sqrt[3]{x} + C$$

Answer: $\frac{3}{4}x\sqrt[3]{x} + C$

💡 Tip: Addition, Subtraction, and Scaling

Integration follows linear properties, allowing you to break down complex expressions:

- **Sum and Difference Rule:** You can integrate terms separately:

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

- **Constant Multiple Rule:** A constant factor can be moved outside the integral:

$$\int k \cdot f(x) dx = k \cdot \int f(x) dx$$

💡 Tip: Combining Constants of Integration

When integrating a function with multiple terms, each term technically produces its own constant (C_1, C_2, C_3 , etc.). However, since the sum or difference of any constant is still just an unknown constant, we always combine them into a single $+C$ at the very end.

$$\int (x + 1) dx = \left(\frac{x^2}{2} + C_1 \right) + (x + C_2) = \frac{x^2}{2} + x + C$$

✔ Solution 5: (c) $\int (t + 1)\sqrt{t} dt$

Rewrite $\sqrt{t}$ as $t^{1/2}$ and distribute into the parentheses:

$$\int (t \cdot t^{1/2} + 1 \cdot t^{1/2}) dt = \int (t^{3/2} + t^{1/2}) dt$$

Integrate term by term using the power rule:

$$\frac{t^{3/2+1}}{3/2+1} + \frac{t^{1/2+1}}{1/2+1} + C = \frac{t^{5/2}}{5/2} + \frac{t^{3/2}}{3/2} + C$$

Simplify the fractions:

$$\frac{2}{5}t^{5/2} + \frac{2}{3}t^{3/2} + C = \frac{2}{5}t^2\sqrt{t} + \frac{2}{3}t\sqrt{t} + C$$

Answer: $\frac{2}{5}t^2\sqrt{t} + \frac{2}{3}t\sqrt{t} + C$

✓ **Solution 5: (d)** $\int (2^x + 3^x) dx$

Apply the exponential rule $\int a^x dx = \frac{a^x}{\ln a} + C$ to each term:

$$\int 2^x dx + \int 3^x dx = \frac{2^x}{\ln 2} + \frac{3^x}{\ln 3} + C$$

Answer: $\frac{2^x}{\ln 2} + \frac{3^x}{\ln 3} + C$

✓ **Solution 5: (e)** $\int \frac{2^x}{3^x} dx$

Rewrite using the property $\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$:

$$\int \left(\frac{2}{3}\right)^x dx$$

Apply the exponential rule with $a = \frac{2}{3}$:

$$\frac{(2/3)^x}{\ln(2/3)} + C = \frac{1}{\ln(2/3)} \cdot \frac{2^x}{3^x} + C$$

Answer: $\frac{1}{\ln(2/3)} \frac{2^x}{3^x} + C$

✓ **Solution 5: (f)** $\int \frac{x^3 + x - 1}{\sqrt[3]{x}} dx$

Divide each term in the numerator by $x^{1/3}$:

$$\int \left(\frac{x^3}{x^{1/3}} + \frac{x^1}{x^{1/3}} - \frac{1}{x^{1/3}} \right) dx = \int (x^{8/3} + x^{2/3} - x^{-1/3}) dx$$

Apply the power rule to each term:

$$\frac{x^{11/3}}{11/3} + \frac{x^{5/3}}{5/3} - \frac{x^{2/3}}{2/3} + C = \frac{3}{11}x^{11/3} + \frac{3}{5}x^{5/3} - \frac{3}{2}x^{2/3} + C$$

Factor out whole powers:

$$\frac{3}{11}x^3 \sqrt[3]{x^2} + \frac{3}{5}x \sqrt[3]{x^2} - \frac{3}{2} \sqrt[3]{x^2} + C$$

Answer: $\frac{3}{11}x^3 \sqrt[3]{x^2} + \frac{3}{5}x \sqrt[3]{x^2} - \frac{3}{2} \sqrt[3]{x^2} + C$

✓ **Solution 5: (g)** $\int (\sqrt{x} + 1)(x - \sqrt{x}) dx$

Expand the expression:

$$\int (\sqrt{x} \cdot x - \sqrt{x} \cdot \sqrt{x} + 1 \cdot x - 1 \cdot \sqrt{x}) dx$$

$$\int (x^{3/2} - x + x - x^{1/2}) dx = \int (x^{3/2} - x^{1/2}) dx$$

Integrate using the power rule:

$$\frac{x^{5/2}}{5/2} - \frac{x^{3/2}}{3/2} + C = \frac{2}{5}x^{5/2} - \frac{2}{3}x^{3/2} + C$$

Answer: $\frac{2}{5}x^2\sqrt{x} - \frac{2}{3}x\sqrt{x} + C$

💡 Tip: Special Cases: Integrals of e^x and Constants

Two fundamental integration rules to remember are:

- **The Exponential Rule:** The integral of e^x is the most straightforward as it remains unchanged:

$$\int e^x dx = e^x + C$$

- **The Constant Rule:** The integral of a constant k (any real number) is that constant multiplied by the variable:

$$\int k dx = kx + C$$

✓ **Solution 5: (h)** $\int \left(x^3 - \frac{7}{x} + e^x \right) dx$

Integrate term by term:

$$\int x^3 dx - 7 \int \frac{1}{x} dx + \int e^x dx$$

Apply the power rule, the log rule, and the exponential rule:

$$\frac{x^4}{4} - 7 \ln |x| + e^x + C$$

Answer: $\frac{x^4}{4} - 7 \ln |x| + e^x + C$

✓ **Solution 5: (i)** $\int \frac{\sqrt{t} - t + t^3}{\sqrt[3]{t}} dt$

Rewrite radicals as fractional powers and divide:

$$\int \left(\frac{t^{1/2}}{t^{1/3}} - \frac{t^1}{t^{1/3}} + \frac{t^3}{t^{1/3}} \right) dt = \int (t^{1/6} - t^{2/3} + t^{8/3}) dt$$

Integrate each term:

$$\frac{t^{7/6}}{7/6} - \frac{t^{5/3}}{5/3} + \frac{t^{11/3}}{11/3} + C = \frac{6}{7}t^{7/6} - \frac{3}{5}t^{5/3} + \frac{3}{11}t^{11/3} + C$$

Answer: $\frac{6}{7}t\sqrt[6]{t} - \frac{3}{5}t\sqrt[3]{t^2} + \frac{3}{11}t^3\sqrt[3]{t^2} + C$

✓ **Solution 5: (j)** $\int (x^e + e^x) dx$

We treat the first term using the Power Rule for a constant exponent ($e \approx 2.718$) and the second term using the Exponential Rule for a constant base:

$$\begin{aligned} \int (x^e + e^x) dx &= \int x^e dx + \int e^x dx \\ &= \frac{x^{e+1}}{e+1} + e^x + C \end{aligned}$$

Answer: $\frac{x^{e+1}}{e+1} + e^x + C$

✓ Solution 5: (k) $\int e^{\sqrt{2}} dx$

Since $e^{\sqrt{2}}$ is a constant (no x variable in the base or exponent), the integral is simply the constant times x :

$$e^{\sqrt{2}} \cdot \int 1 dx = e^{\sqrt{2}}x + C$$

Answer: $e^{\sqrt{2}}x + C$

Question 6

Compute:

a. $\left(\int \frac{x^2 + 7}{x^2 + 10} dx \right)'$

b. $\int \left(\frac{x^2 + 3x - 9}{e^x + \ln x} \right)' dx$

Tip: The "Undo" Trick

For these problems, you don't actually need to do any heavy calculus. You just need to understand that Integration and Derivation are opposites. They "cancel" each other out, like multiplication and division.

Scenario	What's happening?	Result
$(\int f(x) dx)'$	Derivative of an Integral	The original function (no changes)
$\int (f(x))' dx$	Integral of a Derivative	The original function + C

✓ **Solution 6: (a)** $\left(\int \frac{x^2 + 7}{x^2 + 10} dx \right)'$

In this case, we are asked to find the derivative (the prime symbol ' outside the brackets) of an integral.

Think of this as walking forward three steps and then immediately walking backward three steps. You end up exactly where you started.

The derivative "undoes" the integral entirely. We simply remove both symbols and keep the fraction exactly as it is.

Answer: $\frac{x^2 + 7}{x^2 + 10}$

✓ **Solution 6: (b)** $\int \left(\frac{x^2 + 3x - 9}{e^x + \ln x} \right)' dx$

Here, the order is flipped. We are taking the integral of a derivative.

Just like before, the symbols cancel each other out, and we are left with the middle function. However, because the integral symbol is on the outside, we are performing an indefinite calculation.

In business math, whenever the integral is the last thing you do, you must always add the constant $+ C$ at the end to represent any fixed values that might have been lost during derivation.

Answer: $\frac{x^2 + 3x - 9}{e^x + \ln x} + C$

? Question 7

The marginal cost is given by $MC = e^q + q^2$, where q denotes the produced amount. The fixed cost equals 1200 monetary units. Find the total cost function.

💡 Tip: Marginal to Total Cost

Marginal Cost (MC) is the extra cost of producing one more unit. To find the Total Cost (TC), we must "accumulate" those costs by integrating.

- Total Cost (TC) = $\int MC \, dq$
- Fixed Cost = The value of TC when $q = 0$

✓ Solution 7**Step 1: Integrate to find the total cost**

To find the general formula for Total Cost, we integrate the Marginal Cost function with respect to the quantity q :

$$TC = \int (e^q + q^2) \, dq$$

Using our basic integration rules:

$$TC = e^q + \frac{q^3}{3} + C$$

This C represents our potential constant, which we will solve for using the fixed cost.

Step 2: Plug in condition to find C

Fixed cost is the cost when production is zero ($q = 0$). We are told the fixed cost is 1200.

Set $q = 0$ and $TC = 1200$:

$$1200 = e^0 + \frac{0^3}{3} + C$$

Since $e^0 = 1$:

$$1200 = 1 + 0 + C$$

$$C = 1200 - 1 = 1199$$

Now, substitute C back into our equation from Step 1.

Answer: $TC = e^q + \frac{q^3}{3} + 1199$

? Question 8

The marginal revenue is given by $MR = 25 - 20q + 3q^2$, where q denotes the sold amount. Determine the revenue function and the (inverse) demand function.

💡 Tip: Revenue and Demand

Revenue is the total money coming in from sales. The inverse demand function (p) represents the price per unit.

- Total Revenue (TR) = $\int MR dq$
- Revenue at zero sales is zero ($TR = 0$ when $q = 0$)
- Inverse Demand (p) = $\frac{TR}{q}$

✓ Solution 8**Step 1: Integrate to find the revenue function**

To find the Total Revenue, we integrate the Marginal Revenue function:

$$TR = \int (25 - 20q + 3q^2) dq$$

Using our basic integration rules:

$$TR = 25q - \frac{20q^2}{2} + \frac{3q^3}{3} + C$$

$$TR = 25q - 10q^2 + q^3 + C$$

Since revenue must be zero if no items are sold ($q = 0$), then C must be 0.

Step 2: Calculate the inverse demand function

The price p (inverse demand) is simply the Total Revenue divided by the quantity q :

$$p = \frac{25q - 10q^2 + q^3}{q}$$

We divide each term by q to simplify:

$$p = 25 - 10q + q^2$$

Answer: $TR = 25q - 10q^2 + q^3, \quad p = 25 - 10q + q^2$

? Question 9

Find the equation of the curve passing through the point $(3, 2)$ and such that the slope of the tangent line at any of its points (x, y) equals $2x^2 - 5$.

💡 Tip: Tangent Lines and Curves

The slope of a tangent line is simply the derivative of the function. To go from a slope (derivative) back to the original equation of the curve, we use integration.

✓ Solution 9**Step 1: Integrate to find the general equation**

The slope is given by $y' = 2x^2 - 5$. To find y , we integrate this expression with respect to x :

$$y = \int (2x^2 - 5) dx$$

Using the power rule:

$$y = \frac{2x^3}{3} - 5x + C$$

This formula represents a family of curves. We need to find the specific one that passes through the given point.

Step 2: Plug in the point to find C

We are told the curve passes through the point $(3, 2)$. This means that when $x = 3$, y must equal 2.

Substitute these values into our equation:

$$2 = \frac{2(3)^3}{3} - 5(3) + C$$

$$2 = \frac{2(27)}{3} - 15 + C$$

$$2 = 18 - 15 + C$$

$$2 = 3 + C$$

$$C = 2 - 3 = -1$$

Now, substitute $C = -1$ back into the equation from Step 1.

Answer: $y = \frac{2}{3}x^3 - 5x - 1$

Question 10

Find the equation of the curve satisfying $y'' = \frac{24}{x^3}$ and for which the tangent line at the point $(1, 0)$ has equation $y = 12 - 12x$.

Tip: How to find the original curve from y''

When you start with a "double derivative" (y''), you have to work backward in stages to find the original equation (y).

1. Integrate y'' once to get the slope formula (y').
2. Use the slope of the tangent line to find the first missing number (C_1).
3. Integrate y' to get the curve formula (y).
4. Use the point (x, y) to find the second missing number (C_2).

Solution 10**Step 1: Integrate y'' to get the slope formula (y')**

The problem gives us $y'' = 24x^{-3}$. We integrate this to find the formula for the slope:

$$y' = \int 24x^{-3} dx = \frac{24x^{-2}}{-2} + C_1 = -\frac{12}{x^2} + C_1$$

Step 2: Use the tangent line to find C_1

We are told the tangent line is $y = 12 - 12x$. In this line equation, the number in front of x is the slope. So, the slope is -12 at the point where $x = 1$.

$$\begin{aligned} -12 &= -\frac{12}{(1)^2} + C_1 \\ -12 &= -12 + C_1 \implies C_1 = 0 \end{aligned}$$

Our slope formula is now: $y' = -\frac{12}{x^2}$.

Step 3: Integrate y' to get the curve formula (y)

Now we integrate our slope formula to find the actual curve:

$$y = \int -12x^{-2} dx = \frac{-12x^{-1}}{-1} + C_2 = \frac{12}{x} + C_2$$

Step 4: Use the point $(1, 0)$ to find C_2

The curve passes through $(1, 0)$, so when $x = 1$, y must be 0:

$$0 = \frac{12}{1} + C_2$$

$$0 = 12 + C_2 \implies C_2 = -12$$

Answer: $y = \frac{12}{x} - 12$

? Question 11

Use an appropriate substitution such that the integral is reduced to the form $\int u^r du$ and solve:

a. $\int (2x - 7)^5 dx$

b. $\int \frac{1}{\sqrt{5t-2}} dt$

c. $\int x^2 \sqrt{x^3 + 2} dx$

d. $\int \frac{x+3}{\sqrt[3]{x^2+6x}} dx$

💡 Tip: The Substitution Method

When an integral looks too messy, we "swap" the complicated part for a single letter u .

1. Pick the "inside" part to be u .
2. Find du by taking the derivative.
3. Rewrite everything in terms of u , integrate, then swap back to x .

✓ Solution 11: (a) $\int (2x - 7)^5 dx$

We choose the part inside the power to be u so that the whole expression simplifies to u^5 .

$$u = 2x - 7$$

$$du = 2 dx \quad \Rightarrow \quad dx = \frac{du}{2}$$

Substituting these into the integral:

$$\int u^5 \cdot \frac{du}{2} = \frac{1}{2} \int u^5 du = \frac{1}{2} \left(\frac{u^6}{6} \right) + C = \frac{u^6}{12} + C$$

Finally, we replace u with $2x - 7$ to return to the original variable x .

Answer: $\frac{(2x - 7)^6}{12} + C$

✔ **Solution 11: (b)** $\int \frac{1}{\sqrt{5t - 2}} dt$

We choose the expression under the square root to be u to turn the denominator into $\sqrt{u}$.

$$u = 5t - 2$$

$$du = 5 dt \Rightarrow dt = \frac{du}{5}$$

Rewriting the integral using u and $dt = \frac{du}{5}$:

$$\frac{1}{5} \int \frac{1}{\sqrt{u}} du = \frac{1}{5} \int u^{-1/2} du = \frac{1}{5} \left(\frac{u^{1/2}}{1/2} \right) + C = \frac{2\sqrt{u}}{5} + C$$

Finally, we replace u with $5t - 2$ to return to the original variable t .

Answer: $\frac{2\sqrt{5t - 2}}{5} + C$

✓ **Solution 11: (c)** $\int x^2 \sqrt{x^3 + 2} \, dx$

We choose $u = x^3 + 2$ because its derivative involves x^2 , which matches the x^2 already sitting outside the square root.

$$u = x^3 + 2$$

$$du = 3x^2 \, dx \Rightarrow x^2 \, dx = \frac{du}{3}$$

Substituting u for the inside and $\frac{du}{3}$ for the $x^2 \, dx$ part:

$$\int \sqrt{u} \cdot \frac{du}{3} = \frac{1}{3} \int u^{1/2} \, du = \frac{1}{3} \left(\frac{u^{3/2}}{3/2} \right) + C = \frac{2u^{3/2}}{9} + C$$

Finally, we replace u with $x^3 + 2$ to return to the original variable x .

Answer: $\frac{2(x^3 + 2)^{3/2}}{9} + C$

✓ **Solution 11: (d)** $\int \frac{x + 3}{\sqrt[3]{x^2 + 6x}} \, dx$

We choose the denominator's inside part $u = x^2 + 6x$ because its derivative is $2x + 6$, which is exactly twice the $(x + 3)$ on top.

$$u = x^2 + 6x$$

$$du = (2x + 6) \, dx = 2(x + 3) \, dx \Rightarrow (x + 3) \, dx = \frac{du}{2}$$

Replacing the top part with $\frac{du}{2}$ and the bottom with $\sqrt[3]{u}$:

$$\int \frac{1}{\sqrt[3]{u}} \cdot \frac{du}{2} = \frac{1}{2} \int u^{-1/3} \, du = \frac{1}{2} \left(\frac{u^{2/3}}{2/3} \right) + C = \frac{3u^{2/3}}{4} + C$$

Finally, we replace u with $x^2 + 6x$ to return to the original variable x .

Answer: $\frac{3(x^2 + 6x)^{2/3}}{4} + C$

? Question 12

Use an appropriate substitution such that the integral is reduced to the form $\int \frac{1}{u} du$ and solve:

(a) $\int \frac{1}{2x-3} dx$

(b) $\int \frac{x^2}{1-2x^3} dx$

(c) $\int \frac{s+1}{s^2+2s+2} ds$

(d) $\int \frac{e^{2x}}{e^{2x}+3} dx$

(e) $\int \frac{1}{2-x} dx$

💡 Tip: The Substitution Method

When an integral looks too messy, we "swap" the complicated part for a single letter u .

1. Pick the "inside" part to be u .
2. Find du by taking the derivative.
3. Rewrite everything in terms of u , integrate, then swap back to x .

✓ **Solution 12: (a)** $\int \frac{1}{2x-3} dx$

We choose the denominator to be u so that the integral takes the form $1/u$.

$$u = 2x - 3$$

$$du = 2 dx \Rightarrow dx = \frac{du}{2}$$

Substituting these into the integral:

$$\int \frac{1}{u} \cdot \frac{du}{2} = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln |u| + C$$

Finally, we replace u with $2x - 3$ to return to the original variable x .

Answer: $\frac{1}{2} \ln |2x - 3| + C$

✓ **Solution 12: (b)** $\int \frac{x^2}{1-2x^3} dx$

We choose the denominator $u = 1 - 2x^3$ because its derivative involves x^2 , which matches the top of the fraction.

$$u = 1 - 2x^3$$

$$du = -6x^2 dx \Rightarrow x^2 dx = -\frac{du}{6}$$

Substituting into the integral:

$$\int \frac{1}{u} \cdot \left(-\frac{du}{6}\right) = -\frac{1}{6} \int \frac{1}{u} du = -\frac{1}{6} \ln |u| + C$$

Finally, we replace u with $1 - 2x^3$ to return to the original variable x .

Answer: $-\frac{1}{6} \ln |1 - 2x^3| + C$

✓ **Solution 12: (c)** $\int \frac{s+1}{s^2+2s+2} ds$

We choose the denominator $u = s^2 + 2s + 2$. Its derivative is $2s + 2$, which is exactly twice the $(s + 1)$ on top.

$$u = s^2 + 2s + 2$$

$$du = (2s + 2) ds = 2(s + 1) ds \Rightarrow (s + 1) ds = \frac{du}{2}$$

Substituting into the integral:

$$\int \frac{1}{u} \cdot \frac{du}{2} = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln |u| + C$$

Finally, we replace u with $s^2 + 2s + 2$ to return to the original variable s .

Answer: $\frac{1}{2} \ln(s^2 + 2s + 2) + C$

✓ **Solution 12: (d)** $\int \frac{e^{2x}}{e^{2x} + 3} dx$

We choose the denominator $u = e^{2x} + 3$ because its derivative is $2e^{2x}$, matching the numerator.

$$u = e^{2x} + 3$$

$$du = 2e^{2x} dx \Rightarrow e^{2x} dx = \frac{du}{2}$$

Substituting into the integral:

$$\int \frac{1}{u} \cdot \frac{du}{2} = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln |u| + C$$

Finally, we replace u with $e^{2x} + 3$ to return to the original variable x .

Answer: $\frac{1}{2} \ln(e^{2x} + 3) + C$

✔ **Solution 12: (e)** $\int \frac{1}{2-x} dx$

We choose the denominator $u = 2 - x$.

$$u = 2 - x$$

$$du = -1 dx \Rightarrow dx = -du$$

Substituting into the integral:

$$\int \frac{1}{u} \cdot (-du) = -1 \int \frac{1}{u} du = -\ln |u| + C$$

Finally, we replace u with $2 - x$ to return to the original variable x .

Answer: $-\ln |2 - x| + C$

? Question 13

Use an appropriate substitution such that the integral is reduced to the form $\int e^u du$ and solve:

(a) $\int e^{3x} dx$

(b) $\int e^{-x} dx$

(c) $\int \frac{e^{\frac{1}{x}}}{x^2} dx$

💡 Tip: The Substitution Method

When an integral looks too messy, we "swap" the complicated part for a single letter u .

1. Pick the "inside" part to be u .
2. Find du by taking the derivative.
3. Rewrite everything in terms of u , integrate, then swap back to x .

✓ Solution 13: (a) $\int e^{3x} dx$

We choose the exponent to be u so that the integral matches the basic e^u form.

$$u = 3x$$

$$du = 3 dx \quad \Rightarrow \quad dx = \frac{du}{3}$$

Substituting into the integral:

$$\int e^u \cdot \frac{du}{3} = \frac{1}{3} \int e^u du = \frac{1}{3} e^u + C$$

Finally, we replace u with $3x$ to return to the original variable x .

Answer: $\frac{1}{3} e^{3x} + C$

✓ **Solution 13: (b)** $\int e^{-x} dx$

We choose the exponent $u = -x$.

$$u = -x$$

$$du = -1 dx \Rightarrow dx = -du$$

Substituting into the integral:

$$\int e^u \cdot (-du) = -1 \int e^u du = -e^u + C$$

Finally, we replace u with $-x$ to return to the original variable x .

Answer: $-e^{-x} + C$

✓ **Solution 13: (c)** $\int \frac{e^{\frac{1}{x}}}{x^2} dx$

We choose the exponent $u = \frac{1}{x}$ (or x^{-1}) because its derivative is $-\frac{1}{x^2}$, which matches the x^2 in the denominator of the problem.

$$u = \frac{1}{x}$$

$$du = -\frac{1}{x^2} dx \quad \Rightarrow \quad \frac{1}{x^2} dx = -du$$

Substituting u and $-du$ into the integral:

$$\int e^u \cdot (-du) = -\int e^u du = -e^u + C$$

Finally, we replace u with $\frac{1}{x}$ to return to the original variable x .

Answer: $-e^{\frac{1}{x}} + C$

? Question 14

Compute the following integrals using the technique of substitution:

(a) $\int \frac{\sqrt{x} + \ln x}{x} dx$

(f) $\int y^2(1 - y^3)^{\frac{1}{4}} dy$

(b) $\int e^{-2x} dx$

(g) $\int \frac{x + 1}{\sqrt{x^2 + 2x + 5}} dx$

(c) $\int \sqrt{2x + 3} dx$

(h) $\int \frac{5x^2}{(x^3 + 1)^2} dx$

(d) $\int x\sqrt{1 - 3x^2} dx$

(i) $\int \frac{\ln x}{x} dx$

(e) $\int \frac{x^2}{\sqrt[4]{x^3 + 1}} dx$

(j) $\int \frac{x^2}{1 - x^3} dx$

💡 Tip: The Substitution Method

When an integral looks too messy, we "swap" the complicated part for a single letter u . The goal is to turn the integral into a simple form like $\int u^n du$, $\int \frac{1}{u} du$, or $\int e^u du$.

1. Pick the "inside" part to be u .
2. Find du by taking the derivative.
3. Rewrite everything in terms of u , integrate, then swap back to x .

✓ **Solution 14: (a)** $\int \frac{\sqrt{x} + \ln x}{x} dx$

We choose $u = \ln x$ because its derivative $1/x$ is present in the second part of the fraction. We first rewrite the integral as two separate terms:

$$\int \left(\frac{\sqrt{x}}{x} + \frac{\ln x}{x} \right) dx = \int x^{-1/2} dx + \int \frac{\ln x}{x} dx$$

Now we use our substitution for the second term:

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

Substituting u and du into the full expression:

$$\int x^{-1/2} dx + \int u du = 2x^{1/2} + \frac{u^2}{2} + C$$

Finally, we replace u with $\ln x$ to return to the original variable x .

Answer: $2\sqrt{x} + \frac{\ln^2 x}{2} + C$

✓ **Solution 14: (b)** $\int e^{-2x} dx$

We choose the exponent $u = -2x$ to match the standard exponential form.

$$u = -2x$$

$$du = -2 dx \quad \Rightarrow \quad dx = -\frac{du}{2}$$

Substituting these into the integral:

$$\int e^u \cdot \left(-\frac{du}{2}\right) = -\frac{1}{2} \int e^u du = -\frac{1}{2}e^u + C$$

Finally, we replace u with $-2x$ to return to the original variable x .

Answer: $-\frac{1}{2}e^{-2x} + C$

✓ **Solution 14: (c)** $\int \sqrt{2x+3} dx$

We choose $u = 2x + 3$ to simplify the expression under the root.

$$u = 2x + 3$$

$$du = 2 dx \Rightarrow dx = \frac{du}{2}$$

Substituting into the integral and using $u^{1/2}$ for the square root:

$$\int u^{1/2} \cdot \frac{du}{2} = \frac{1}{2} \left(\frac{u^{3/2}}{3/2} \right) + C = \frac{1}{3}u^{3/2} + C$$

Finally, we replace u with $2x + 3$ to return to the original variable x .

Answer: $\frac{1}{3}\sqrt{(2x+3)^3} + C$

✓ **Solution 14: (d)** $\int x \sqrt{1 - 3x^2} dx$

We choose $u = 1 - 3x^2$ because its derivative $-6x$ matches the x outside the root.

$$u = 1 - 3x^2$$

$$du = -6x dx \Rightarrow x dx = -\frac{du}{6}$$

Substituting these into the integral:

$$\int \sqrt{u} \cdot \left(-\frac{du}{6}\right) = -\frac{1}{6} \int u^{1/2} du = -\frac{1}{6} \left(\frac{u^{3/2}}{3/2}\right) + C$$

Finally, we replace u with $1 - 3x^2$ to return to the original variable x .

Answer: $-\frac{1}{9} \sqrt{(1 - 3x^2)^3} + C$

✓ **Solution 14: (e)** $\int \frac{x^2}{\sqrt[4]{x^3 + 1}} dx$

We choose $u = x^3 + 1$ because its derivative $3x^2$ matches the x^2 on top.

$$u = x^3 + 1$$

$$du = 3x^2 dx \Rightarrow x^2 dx = \frac{du}{3}$$

Substituting into the integral and using $u^{-1/4}$ for the root on the bottom:

$$\int \frac{1}{u^{1/4}} \cdot \frac{du}{3} = \frac{1}{3} \int u^{-1/4} du = \frac{1}{3} \left(\frac{u^{3/4}}{3/4}\right) + C$$

Finally, we replace u with $x^3 + 1$ to return to the original variable x .

Answer: $\frac{4}{9} \sqrt[4]{(x^3 + 1)^3} + C$

✓ **Solution 14: (f)** $\int y^2(1 - y^3)^{\frac{1}{4}} dy$

We choose $u = 1 - y^3$ because its derivative $du = -3y^2 dy$ contains the y^2 term that is outside the parentheses.

$$u = 1 - y^3$$

$$du = -3y^2 dy \Rightarrow y^2 dy = -\frac{du}{3}$$

Substituting these into the integral to turn it into a power rule problem:

$$\int u^{1/4} \cdot \left(-\frac{du}{3}\right) = -\frac{1}{3} \int u^{1/4} du = -\frac{1}{3} \left(\frac{u^{5/4}}{5/4}\right) + C$$

Finally, we replace u with $1 - y^3$ to return to the original variable y .

Answer: $-\frac{4}{15}(1 - y^3)^{\frac{5}{4}} + C$

✓ **Solution 14: (g)** $\int \frac{x + 1}{\sqrt{x^2 + 2x + 5}} dx$

We choose $u = x^2 + 2x + 5$ because the derivative $2x + 2$ is exactly double the numerator

$(x + 1)$.

$$u = x^2 + 2x + 5$$

$$du = (2x + 2) dx = 2(x + 1) dx \Rightarrow (x + 1) dx = \frac{du}{2}$$

Substituting into the integral and rewriting the root as a negative power:

$$\int \frac{1}{\sqrt{u}} \cdot \frac{du}{2} = \frac{1}{2} \int u^{-1/2} du = \frac{1}{2} \left(\frac{u^{1/2}}{1/2} \right) + C$$

Finally, we replace u with $x^2 + 2x + 5$ to return to the original variable x .

Answer: $\sqrt{x^2 + 2x + 5} + C$

✓ **Solution 14: (h)** $\int \frac{5x^2}{(x^3 + 1)^2} dx$

We choose $u = x^3 + 1$ because its derivative $3x^2$ matches the x^2 on top. We move the constant 5 outside to simplify.

$$u = x^3 + 1$$

$$du = 3x^2 dx \Rightarrow x^2 dx = \frac{du}{3}$$

Substituting into the integral:

$$5 \int \frac{1}{u^2} \cdot \frac{du}{3} = \frac{5}{3} \int u^{-2} du = \frac{5}{3} \left(\frac{u^{-1}}{-1} \right) + C$$

Finally, we replace u with $x^3 + 1$ to return to the original variable x .

Answer: $-\frac{5}{3(x^3 + 1)} + C$

✓ **Solution 14: (i)** $\int \frac{\ln x}{x} dx$

We choose $u = \ln x$ because the derivative $1/x$ is multiplying the function.

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

Substituting u and du into the integral:

$$\int u du = \frac{u^2}{2} + C$$

Finally, we replace u with $\ln x$ to return to the original variable x .

Answer: $\frac{1}{2} \ln^2 x + C$

✓ **Solution 14: (j)** $\int \frac{x^2}{1 - x^3} dx$

We choose the denominator $u = 1 - x^3$ because its derivative involves x^2 , leading to a natural log form.

$$u = 1 - x^3$$

$$du = -3x^2 dx \Rightarrow x^2 dx = -\frac{du}{3}$$

Substituting into the integral:

$$\int \frac{1}{u} \cdot \left(-\frac{du}{3}\right) = -\frac{1}{3} \int \frac{1}{u} du = -\frac{1}{3} \ln |u| + C$$

Finally, we replace u with $1 - x^3$ to return to the original variable x .

Answer: $-\frac{1}{3} \ln |1 - x^3| + C$

Question 15

The marginal cost is given by $MC = 20 + 12q(q^2 + 2)^2$ and the marginal revenue is given by $MR = \frac{4}{(1+q)^2} + \frac{2}{1+q} + 3$. The total cost for a production of 3 units amounts to 2760 monetary units.

- Determine the total cost function.
- Determine the average cost function.
- Determine the revenue function.
- Determine the (inverse) demand function.

Tip: Economic Functions

To move from marginal functions to total functions, we integrate. To find the specific constant C , we use the provided initial conditions (fixed points).

- Total Cost:** $TC(q) = \int MC \, dq$
- Average Cost:** $AC(q) = \frac{TC(q)}{q}$
- Total Revenue:** $TR(q) = \int MR \, dq$ (Note: $TR(0) = 0$)
- Demand (Price):** $p = \frac{TR(q)}{q}$

Solution 15: (a) Determine the total cost function

We integrate the marginal cost function $MC = 20 + 12q(q^2 + 2)^2$:

$$TC(q) = \int [20 + 12q(q^2 + 2)^2] \, dq = \int 20 \, dq + \int 12q(q^2 + 2)^2 \, dq$$

For the second part, we use u -substitution:

$$u = q^2 + 2$$

$$du = 2q \, dq \Rightarrow 12q \, dq = 6 \, du$$

Substituting into the integral:

$$TC(q) = 20q + \int 6u^2 \, du = 20q + 2u^3 + C = 20q + 2(q^2 + 2)^3 + C$$

Using the condition that $TC(3) = 2760$:

$$2760 = 20(3) + 2(3^2 + 2)^3 + C = 60 + 2(11)^3 + C = 60 + 2662 + C$$

$$2760 = 2722 + C \Rightarrow C = 38$$

Answer: $TC(q) = 20q + 2(q^2 + 2)^3 + 38$

✓ Solution 15: (b) Determine the average cost function

The average cost is the total cost divided by the quantity q :

$$AC(q) = \frac{TC(q)}{q} = \frac{20q + 2(q^2 + 2)^3 + 38}{q}$$

Distributing the q through the terms:

Answer: $AC(q) = 20 + \frac{2(q^2+2)^3}{q} + \frac{38}{q}$

✓ **Solution 15: (c) Determine the revenue function**

We integrate the marginal revenue function $MR = \frac{4}{(1+q)^2} + \frac{2}{1+q} + 3$:

$$TR(q) = \int \left[4(1+q)^{-2} + 2(1+q)^{-1} + 3 \right] dq$$

To integrate the first two terms, we use the substitution:

$$u = 1 + q \quad \Rightarrow \quad du = dq$$

Applying the power rule and the natural logarithm rule:

$$\begin{aligned} TR(q) &= \int 4u^{-2} du + \int \frac{2}{u} du + \int 3 dq \\ &= 4 \left(\frac{u^{-1}}{-1} \right) + 2 \ln |u| + 3q + C \\ &= -\frac{4}{1+q} + 2 \ln |1+q| + 3q + C \end{aligned}$$

Since total revenue must be zero when no units are sold ($TR(0) = 0$), we solve for the constant C :

$$\begin{aligned} 0 &= -\frac{4}{1+0} + 2 \ln |1+0| + 3(0) + C \\ 0 &= -4 + 0 + 0 + C \quad \Rightarrow \quad C = 4 \end{aligned}$$

Substituting $C = 4$ back into the function:

$$TR(q) = 2 \ln(1+q) + 3q + 4 - \frac{4}{1+q}$$

To simplify, we combine the algebraic terms over the common denominator $(1 + q)$:

$$\begin{aligned} TR(q) &= 2 \ln(1 + q) + \frac{(3q + 4)(1 + q) - 4}{1 + q} \\ &= 2 \ln(1 + q) + \frac{3q^2 + 3q + 4q + 4 - 4}{1 + q} \\ &= 2 \ln(1 + q) + \frac{3q^2 + 7q}{1 + q} \end{aligned}$$

Answer: $TR(q) = 2 \ln(1 + q) + \frac{3q^2 + 7q}{1 + q}$

✓ Solution 15: (d) Determine the (inverse) demand function

The demand function p is the total revenue divided by q :

$$p = \frac{TR(q)}{q} = \frac{2 \ln(1 + q) + \frac{3q^2 + 7q}{1 + q}}{q}$$

Dividing each part of the numerator by q :

$$p = \frac{2 \ln(1 + q)}{q} + \frac{q(3q + 7)}{q(1 + q)}$$

Answer: $p = \frac{2 \ln(1 + q)}{q} + \frac{3q + 7}{1 + q}$

Question 16

Compute using the technique of integration by parts:

(a) $\int x^2 e^x dx$

(f) $\int x \ln x dx$

(k) $\int x \sqrt{x-2} dx$

(b) $\int x^2 \ln x dx$

(g) $\int (\ln x)^2 dx$

(l) $\int \frac{\ln x}{x^3} dx$

(c) $\int x e^{-x} dx$

(h) $\int (e^x + x)^2 dx$

(m) $\int \frac{v}{\sqrt{v+3}} dv$

(d) $\int x \sqrt{1+x} dx$

(i) $\int x^3 e^{x^2} dx$

(n) $\int x^r \ln x dx \quad (r \neq -1)$

(e) $\int x^2 \sqrt{x+5} dx$

(j) $\int x^2 e^{-3x} dx$

(o) $\int x^3 e^{2x} dx$

Hints:

- For (g), take $f(x) = (\ln x)^2$ and $g'(x) = 1$.
- For (h), work out the brackets first by using the formula for $(a+b)^2$.
- For (i), write x^3 as $x^2 \cdot x$ and take $x e^{x^2}$ as $g'(x)$.

Tip: Integration by Parts Formula

Instead of substitution, we use this formula for products of different function types:

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$

We organize our terms in this specific layout:

$$f(x) \Rightarrow f'(x)$$

$$g'(x) \Rightarrow g(x)$$

💡 Tip: Choosing the Derivative Term (The LAE Rule)

When using integration by parts, we use the LAE priority list to decide which part of the integrand should be our function $f(x)$ (the part we differentiate). The first type of function you see in this list becomes $f(x)$:

1. **L** – Logarithmic functions (e.g., $\ln x$)
2. **A** – Algebraic functions (e.g., x^n , $\sqrt{x}$)
3. **E** – Exponential functions (e.g., e^x , 2^x)

✔ Solution 16: (a) $\int x^2 e^x dx$

The integrand contains an Algebraic term (x^2) and an Exponential term (e^x). According to LAE, Algebraic comes before Exponential, so we differentiate x^2 .

$$f(x) = x^2 \Rightarrow f'(x) = 2x$$

$$g'(x) = e^x \Rightarrow g(x) = e^x$$

Applying the formula:

$$\int x^2 e^x dx = x^2 e^x - \int 2x e^x dx$$

We must apply the technique again to the remaining integral $\int 2x e^x dx$. Following the same

LAE logic, we keep the Algebraic term as $f(x)$ and the Exponential term as $g'(x)$:

$$f(x) = 2x \Rightarrow f'(x) = 2$$

$$g'(x) = e^x \Rightarrow g(x) = e^x$$

$$\int x^2 e^x dx = x^2 e^x - \left(2x e^x - \int 2e^x dx \right) = x^2 e^x - 2x e^x + 2e^x + C$$

Answer: $(x^2 - 2x + 2)e^x + C$

✓ Solution 16: (b) $\int x^2 \ln x dx$

The integrand contains an Algebraic term (x^2) and a Logarithmic term ($\ln x$). According to LAE, Logarithmic comes first, so we differentiate $\ln x$.

$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$$

$$g'(x) = x^2 \Rightarrow g(x) = \frac{x^3}{3}$$

Applying the formula:

$$\int x^2 \ln x dx = (\ln x) \left(\frac{x^3}{3} \right) - \int \frac{1}{x} \cdot \frac{x^3}{3} dx = \frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^2 dx$$

$$\int x^2 \ln x dx = \frac{1}{3} x^3 \ln x - \frac{1}{3} \left(\frac{x^3}{3} \right) + C$$

Answer: $\frac{1}{3} x^3 \ln x - \frac{1}{9} x^3 + C$

✓ Solution 16: (c) $\int x e^{-x} dx$

We have an Algebraic term (x) and an Exponential term (e^{-x}). LAE dictates that Algebraic takes priority for differentiation.

$$f(x) = x \Rightarrow f'(x) = 1$$

$$g'(x) = e^{-x} \Rightarrow g(x) = -e^{-x}$$

Applying the formula:

$$\int x e^{-x} dx = x(-e^{-x}) - \int 1 \cdot (-e^{-x}) dx = -x e^{-x} + \int e^{-x} dx$$

$$\int x e^{-x} dx = -x e^{-x} - e^{-x} + C$$

Answer: $-(x + 1)e^{-x} + C$

✓ Solution 16: (d) $\int x \sqrt{1+x} dx$

This is composed of two Algebraic parts. We choose $f(x) = x$ because its derivative is 1, which simplifies the integral significantly compared to differentiating the root.

$$f(x) = x \Rightarrow f'(x) = 1$$

$$g'(x) = (1+x)^{1/2} \Rightarrow g(x) = \frac{2}{3}(1+x)^{3/2}$$

Applying the formula:

$$\int x(1+x)^{1/2} dx = \frac{2}{3}x(1+x)^{3/2} - \int 1 \cdot \frac{2}{3}(1+x)^{3/2} dx$$

$$\int x(1+x)^{1/2} dx = \frac{2}{3}x(1+x)^{3/2} - \frac{2}{3} \left(\frac{(1+x)^{5/2}}{5/2} \right) + C$$

Answer: $\frac{2}{3}x\sqrt{(x+1)^3} - \frac{4}{15}\sqrt{(x+1)^5} + C$

✓ Solution 16: (e) $\int x^2 \sqrt{x+5} dx$

The integrand contains two Algebraic terms (x^2 and $\sqrt{x+5}$). We choose to differentiate x^2 because its degree reduces with each step. Based on the LAE rule, we treat x^2 as our $f(x)$:

$$f(x) = x^2 \Rightarrow f'(x) = 2x$$

$$g'(x) = (x+5)^{1/2} \Rightarrow g(x) = \frac{2}{3}(x+5)^{3/2}$$

Applying the formula $\int f g' = f g - \int f' g$:

$$\int x^2 \sqrt{x+5} dx = \frac{2}{3}x^2(x+5)^{3/2} - \int \frac{4}{3}x(x+5)^{3/2} dx$$

We now apply the technique to the remaining integral $\int \frac{4}{3}x(x+5)^{3/2} dx$. Again, following the LAE priority, we differentiate the linear term:

$$f(x) = \frac{4}{3}x \Rightarrow f'(x) = \frac{4}{3}$$

$$g'(x) = (x+5)^{3/2} \Rightarrow g(x) = \frac{2}{5}(x+5)^{5/2}$$

Substituting these results back into the equation and distributing the negative sign:

$$\begin{aligned}
 \int x^2 \sqrt{x+5} \, dx &= \frac{2}{3}x^2(x+5)^{3/2} - \left(\frac{8}{15}x(x+5)^{5/2} - \int \frac{4}{3} \cdot \frac{2}{5}(x+5)^{5/2} \, dx \right) \\
 &= \frac{2}{3}x^2(x+5)^{3/2} - \frac{8}{15}x(x+5)^{5/2} + \int \frac{8}{15}(x+5)^{5/2} \, dx \\
 &= \frac{2}{3}x^2(x+5)^{3/2} - \frac{8}{15}x(x+5)^{5/2} + \frac{8}{15} \left(\frac{(x+5)^{7/2}}{7/2} \right) + C \\
 &= \frac{2}{3}x^2(x+5)^{3/2} - \frac{8}{15}x(x+5)^{5/2} + \frac{16}{105}(x+5)^{7/2} + C
 \end{aligned}$$

Answer: $\frac{2}{3}x^2\sqrt{(x+5)^3} - \frac{8}{15}x\sqrt{(x+5)^5} + \frac{16}{105}\sqrt{(x+5)^7} + C$

✓ Solution 16: (f) $\int x \ln x \, dx$

The integrand contains an Algebraic term (x) and a Logarithmic term ($\ln x$). Based on the LAE rule, Logarithmic functions take priority for differentiation over Algebraic ones.

$$f(x) = \ln x \quad \Rightarrow \quad f'(x) = \frac{1}{x}$$

$$g'(x) = x \quad \Rightarrow \quad g(x) = \frac{x^2}{2}$$

Applying the integration by parts formula $\int f(x)g'(x) \, dx = f(x)g(x) - \int f'(x)g(x) \, dx$:

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{1}{x} \cdot \frac{x^2}{2} \, dx = \frac{1}{2}x^2 \ln x - \frac{1}{2} \int x \, dx$$

Performing the final integration:

$$\int x \ln x \, dx = \frac{1}{2}x^2 \ln x - \frac{1}{2} \left(\frac{x^2}{2} \right) + C$$

Answer: $\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2 + C$

✓ **Solution 16: (g)** $\int (\ln x)^2 dx$

We have a Logarithmic term $(\ln x)^2$. Following the hint and the LAE rule, we differentiate the Logarithmic part and treat the invisible 1 as the Algebraic part $g'(x)$.

$$f(x) = (\ln x)^2 \Rightarrow f'(x) = \frac{2 \ln x}{x}$$

$$g'(x) = 1 \Rightarrow g(x) = x$$

Applying the formula:

$$\int (\ln x)^2 dx = x(\ln x)^2 - \int \frac{2 \ln x}{x} \cdot x dx = x(\ln x)^2 - 2 \int \ln x dx$$

We must use integration by parts again for $\int \ln x dx$:

$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$$

$$g'(x) = 1 \Rightarrow g(x) = x$$

The full expression becomes:

$$x(\ln x)^2 - 2 \left(x \ln x - \int \frac{1}{x} \cdot x dx \right) = x(\ln x)^2 - 2x \ln x + 2x + C$$

Answer: $x(\ln x)^2 - 2x \ln x + 2x + C$

✔ **Solution 16: (h)** $\int (e^x + x)^2 dx$

As suggested in the hint, we first expand the Algebraic bracket using $(a + b)^2 = a^2 + 2ab + b^2$:

$$\int (e^x + x)^2 dx = \int (e^{2x} + 2xe^x + x^2) dx = \int e^{2x} dx + \int 2xe^x dx + \int x^2 dx$$

The middle term $\int 2xe^x dx$ requires integration by parts. With an Algebraic term ($2x$) and Exponential (e^x), LAE gives priority to $2x$.

$$f(x) = 2x \Rightarrow f'(x) = 2$$

$$g'(x) = e^x \Rightarrow g(x) = e^x$$

Solving the three separate integrals:

$$\frac{1}{2}e^{2x} + \left(2xe^x - \int 2e^x dx\right) + \frac{x^3}{3} = \frac{1}{2}e^{2x} + 2xe^x - 2e^x + \frac{1}{3}x^3 + C$$

Answer: $\frac{1}{2}e^{2x} + 2xe^x - 2e^x + \frac{1}{3}x^3 + C$

✔ **Solution 16: (i)** $\int x^3 e^{x^2} dx$

Following the hint, we rewrite the integrand as $x^2 \cdot (xe^{x^2})$ to facilitate integration by parts. According to the LAE rule, we choose the algebraic term x^2 to differentiate:

$$f(x) = x^2 \Rightarrow f'(x) = 2x$$

$$g'(x) = xe^{x^2} \Rightarrow g(x) = \int xe^{x^2} dx$$

To find $g(x)$, we use u -substitution for the integral $\int x e^{x^2} dx$:

$$\text{Let } u = x^2 \quad \Rightarrow \quad du = 2x dx \quad \Rightarrow \quad \frac{1}{2} du = x dx$$

$$\begin{aligned} g(x) &= \int e^u \cdot \left(\frac{1}{2}\right) du \\ &= \frac{1}{2} e^u = \frac{1}{2} e^{x^2} \end{aligned}$$

Finally, using the integration by parts formula:

$$\begin{aligned} \int x^3 e^{x^2} dx &= (x^2) \left(\frac{1}{2} e^{x^2}\right) - \int (2x) \left(\frac{1}{2} e^{x^2}\right) dx \\ &= \frac{1}{2} x^2 e^{x^2} - \int x e^{x^2} dx \end{aligned}$$

As we already determined the integral of $x e^{x^2}$:

$$\int x^3 e^{x^2} dx = \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + C$$

Answer: $\frac{1}{2} e^{x^2} (x^2 - 1) + C$

✓ Solution 16: (j) $\int x^2 e^{-3x} dx$

We have an Algebraic term (x^2) and an Exponential term (e^{-3x}). LAE dictates we differentiate the Algebraic term.

$$f(x) = x^2 \quad \Rightarrow \quad f'(x) = 2x$$

$$g'(x) = e^{-3x} \quad \Rightarrow \quad g(x) = -\frac{1}{3} e^{-3x}$$

First application:

$$-\frac{1}{3}x^2e^{-3x} - \int 2x \cdot \left(-\frac{1}{3}e^{-3x}\right) dx = -\frac{1}{3}x^2e^{-3x} + \frac{2}{3} \int xe^{-3x} dx$$

Second application for $\int xe^{-3x} dx$ with $f(x) = x$ and $g'(x) = e^{-3x}$:

$$f(x) = x \Rightarrow f'(x) = 1$$

$$g'(x) = e^{-3x} \Rightarrow g(x) = -\frac{1}{3}e^{-3x}$$

Substituting back:

$$-\frac{1}{3}x^2e^{-3x} + \frac{2}{3} \left(-\frac{1}{3}xe^{-3x} - \int -\frac{1}{3}e^{-3x} dx \right) = -\frac{1}{3}x^2e^{-3x} - \frac{2}{9}xe^{-3x} - \frac{2}{27}e^{-3x} + C$$

Answer: $-\frac{1}{3}x^2e^{-3x} - \frac{2}{9}xe^{-3x} - \frac{2}{27}e^{-3x} + C$

✓ Solution 16: (k) $\int x\sqrt{x-2} dx$

This integral consists of two Algebraic functions. Following the LAE priority, we choose the simpler Algebraic term $f(x) = x$ to differentiate so that it becomes a constant, while $g'(x)$ handles the square root.

$$f(x) = x \Rightarrow f'(x) = 1$$

$$g'(x) = (x-2)^{1/2} \Rightarrow g(x) = \frac{2}{3}(x-2)^{3/2}$$

Applying the integration by parts formula:

$$\int x(x-2)^{1/2} dx = \frac{2}{3}x(x-2)^{3/2} - \int 1 \cdot \frac{2}{3}(x-2)^{3/2} dx$$

Performing the final integration on the remaining term:

$$\int x(x-2)^{1/2} dx = \frac{2}{3}x(x-2)^{3/2} - \frac{2}{3} \left(\frac{(x-2)^{5/2}}{5/2} \right) + C$$

Answer: $\frac{2}{3}x\sqrt{(x-2)^3} - \frac{4}{15}\sqrt{(x-2)^5} + C$

✓ **Solution 16: (I)** $\int \frac{\ln x}{x^3} dx$

The integrand contains a Logarithmic term ($\ln x$) and an Algebraic term (x^{-3}). According to the LAE rule, the Logarithmic term takes priority for differentiation.

$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$$

$$g'(x) = x^{-3} \Rightarrow g(x) = -\frac{1}{2x^2}$$

Applying the formula:

$$\int (\ln x)x^{-3} dx = -\frac{\ln x}{2x^2} - \int \left(\frac{1}{x}\right) \left(-\frac{1}{2x^2}\right) dx = -\frac{\ln x}{2x^2} + \frac{1}{2} \int x^{-3} dx$$

Performing the final integration:

$$\int \frac{\ln x}{x^3} dx = -\frac{\ln x}{2x^2} + \frac{1}{2} \left(-\frac{1}{2x^2} \right) + C$$

Answer: $-\frac{1}{2} \frac{\ln x}{x^2} - \frac{1}{4x^2} + C$

✓ **Solution 16: (m)** $\int \frac{v}{\sqrt{v+3}} dv$

We treat both parts as Algebraic. We choose $f(v) = v$ to differentiate because its derivative is 1, simplifying the resulting integral.

$$f(v) = v \Rightarrow f'(v) = 1$$

$$g'(v) = (v+3)^{-1/2} \Rightarrow g(v) = 2(v+3)^{1/2}$$

Applying the formula:

$$\int v(v+3)^{-1/2} dv = 2v(v+3)^{1/2} - \int 1 \cdot 2(v+3)^{1/2} dv$$

Performing the final integration:

$$\int v(v+3)^{-1/2} dv = 2v\sqrt{v+3} - 2 \left(\frac{(v+3)^{3/2}}{3/2} \right) + C$$

Answer: $2v\sqrt{v+3} - \frac{4}{3}\sqrt{(v+3)^3} + C$

✓ **Solution 16: (n)** $\int x^r \ln x dx \quad (r \neq -1)$

Following the LAE rule, the Logarithmic term $\ln x$ is chosen for differentiation over the Algebraic term x^r .

$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$$

$$g'(x) = x^r \Rightarrow g(x) = \frac{x^{r+1}}{r+1}$$

Applying the formula:

$$\int (\ln x)x^r dx = \frac{x^{r+1}}{r+1} \ln x - \int \frac{1}{x} \cdot \frac{x^{r+1}}{r+1} dx = \frac{x^{r+1}}{r+1} \ln x - \frac{1}{r+1} \int x^r dx$$

Performing the final integration:

$$\int x^r \ln x \, dx = \frac{x^{r+1}}{r+1} \ln x - \frac{1}{r+1} \left(\frac{x^{r+1}}{r+1} \right) + C$$

Answer: $\frac{1}{r+1} x^{r+1} \ln x - \frac{x^{r+1}}{(r+1)^2} + C$

✔ **Solution 16: (o)** $\int x^3 e^{2x} \, dx$

The integrand consists of an Algebraic term (x^3) and an Exponential term (e^{2x}). Following the LAE rule, we differentiate the Algebraic terms:

$$f(x) = x^3 \Rightarrow f'(x) = 3x^2$$

$$g'(x) = e^{2x} \Rightarrow g(x) = \frac{1}{2} e^{2x}$$

Applying the integration by parts formula:

$$\int x^3 e^{2x} \, dx = \frac{1}{2} x^3 e^{2x} - \int \frac{3}{2} x^2 e^{2x} \, dx$$

Applying the technique to $\int \frac{3}{2} x^2 e^{2x} \, dx$:

$$f(x) = \frac{3}{2} x^2 \Rightarrow f'(x) = 3x$$

$$g'(x) = e^{2x} \Rightarrow g(x) = \frac{1}{2} e^{2x}$$

Substituting this back into the expression:

$$\begin{aligned} \int x^3 e^{2x} \, dx &= \frac{1}{2} x^3 e^{2x} - \left(\frac{3}{4} x^2 e^{2x} - \int \frac{3}{2} x e^{2x} \, dx \right) \\ &= \frac{1}{2} x^3 e^{2x} - \frac{3}{4} x^2 e^{2x} + \int \frac{3}{2} x e^{2x} \, dx \end{aligned}$$

Applying the technique to $\int \frac{3}{2}x e^{2x} dx$:

$$f(x) = \frac{3}{2}x \Rightarrow f'(x) = \frac{3}{2}$$

$$g'(x) = e^{2x} \Rightarrow g(x) = \frac{1}{2}e^{2x}$$

Substituting and completing the integration:

$$\begin{aligned}\int x^3 e^{2x} dx &= \frac{1}{2}x^3 e^{2x} - \frac{3}{4}x^2 e^{2x} + \left(\frac{3}{4}x e^{2x} - \int \frac{3}{4}e^{2x} dx \right) \\ &= \frac{1}{2}x^3 e^{2x} - \frac{3}{4}x^2 e^{2x} + \frac{3}{4}x e^{2x} - \frac{3}{8}e^{2x} + C\end{aligned}$$

Answer: $\left(\frac{1}{2}x^3 - \frac{3}{4}x^2 + \frac{3}{4}x - \frac{3}{8} \right) e^{2x} + C$

? Question 17

Write in partial fractions. Check your result by rewriting your solution as one single fraction.

(a) $\frac{1}{x^2 - 9x + 20}$

(c) $\frac{3}{2 - x^2}$

(b) $\frac{4}{x^2 - 9}$

(d) $\frac{1}{x - 0.01x^2}$

💡 Tip: Partial Fraction Decomposition Rules

For a rational function where the denominator is a quadratic (degree 2) and the numerator is a constant or linear (degree 0 or 1), we use the following decomposition types based on the factors of the denominator:

- **Case 1: Distinct Linear Factors**

$$\frac{1}{(x - a)(x - b)} = \frac{A}{x - a} + \frac{B}{x - b}$$

- **Case 2: Repeated Linear Factors**

$$\frac{1}{(x - a)^2} = \frac{A}{x - a} + \frac{B}{(x - a)^2}$$

💡 Tip: The 6-Step Program for Partial Fractions

To decompose a rational function with a quadratic denominator, follow these systematic steps:

1. **Find the zeros** of the denominator by setting it equal to zero.
2. **Factor the denominator** using the zeros found in Step 1.
3. **Write the setup** with unknown constant numerators (A, B) over each factor.

4. **Sum the partial fractions** by finding a common denominator to create a single expression.
5. **Equal the numerators** of your new expression to the original numerator to form a system of equations.
6. **Solve for the unknowns** (A and B) to complete the decomposition.

✓ **Solution 17: (a)** $\frac{1}{x^2 - 9x + 20}$

Step 1: Find the zeros of the denominator

We set the quadratic denominator to zero to find the roots of the expression:

$$x^2 - 9x + 20 = 0$$

Using the quadratic formula or factoring, we find the values that satisfy this:

$$(x - 4)(x - 5) = 0$$

The zeros are $x = 4$ and $x = 5$.

Step 2: Factor the denominator

Using the zeros found in Step 1, we rewrite the denominator in its linear factored form:

$$x^2 - 9x + 20 = (x - 5)(x - 4)$$

Step 3: Write the partial fractions with unknown constant numerators

Since we have two distinct linear factors, we assign a constant (A and B) to each factor:

$$\frac{1}{(x - 5)(x - 4)} = \frac{A}{x - 5} + \frac{B}{x - 4}$$

Step 4: Sum the partial fractions

To combine these, we find a common denominator by multiplying each term by the other's denominator:

$$\frac{A(x - 4) + B(x - 5)}{(x - 5)(x - 4)}$$

Expanding and grouping the terms in the numerator by the variable x :

$$\frac{Ax - 4A + Bx - 5B}{x^2 - 9x + 20} = \frac{(A + B)x + (-4A - 5B)}{x^2 - 9x + 20}$$

Step 5: Equal the numerators

We equate our new numerator to the original numerator, which is the constant 1. Since there is no x term in the original, its coefficient must be 0:

$$(A + B)x + (-4A - 5B) = 0x + 1$$

This creates the following system of equations:

$$\begin{cases} A + B = 0 \\ -4A - 5B = 1 \end{cases}$$

Step 6: Solve for the unknowns

From the first equation, we see that $A = -B$. Substituting this into the second equation:

$$-4(-B) - 5B = 1$$

$$4B - 5B = 1 \Rightarrow -B = 1 \Rightarrow B = -1$$

Substituting $B = -1$ back into our expression for A :

$$A = -(-1) \Rightarrow A = 1$$

Answer: $\frac{1}{x - 5} - \frac{1}{x - 4}$

✓ Solution 17: (b) $\frac{4}{x^2 - 9}$

Step 1: Find the zeros of the denominator

We set the quadratic denominator equal to zero to find the roots:

$$x^2 - 9 = 0$$

Solving for x :

$$x^2 = 9 \Rightarrow x = \pm 3$$

The zeros are $x = 3$ and $x = -3$.

Step 2: Factor the denominator

Using the zeros from Step 1 or the difference of squares identity $a^2 - b^2 = (a - b)(a + b)$:

$$x^2 - 9 = (x - 3)(x + 3)$$

Step 3: Write the partial fractions with unknown constant numerators

We assign a constant to each distinct linear factor:

$$\frac{4}{(x - 3)(x + 3)} = \frac{A}{x - 3} + \frac{B}{x + 3}$$

Step 4: Sum the partial fractions

Combine the fractions by finding a common denominator:

$$\frac{A(x + 3) + B(x - 3)}{(x - 3)(x + 3)}$$

Expanding the numerator and grouping the terms by the variable x :

$$\frac{Ax + 3A + Bx - 3B}{x^2 - 9} = \frac{(A + B)x + (3A - 3B)}{x^2 - 9}$$

Step 5: Equal the numerators

We equate our new expression to the original numerator, 4. Since there is no x term in the original, its coefficient is 0:

$$(A + B)x + (3A - 3B) = 0x + 4$$

This gives us the following system of equations:

$$\begin{cases} A + B = 0 \\ 3A - 3B = 4 \end{cases}$$

Step 6: Solve for the unknowns

From the first equation, $A = -B$. Substituting this into the second equation:

$$\begin{aligned} 3(-B) - 3B &= 4 \\ -6B &= 4 \Rightarrow B = -\frac{4}{6} = -\frac{2}{3} \end{aligned}$$

Now find A using the relationship $A = -B$:

$$A = -\left(-\frac{2}{3}\right) = \frac{2}{3}$$

Answer: $\frac{2}{3(x-3)} - \frac{2}{3(x+3)}$

✓ **Solution 17: (c)** $\frac{3}{2-x^2}$

Step 1: Find the zeros of the denominator

We determine where the quadratic expression in the denominator equals zero:

$$2 - x^2 = 0$$

Solving for x :

$$x^2 = 2 \Rightarrow x = \pm\sqrt{2}$$

The zeros are $x = \sqrt{2}$ and $x = -\sqrt{2}$.

Step 2: Factor the denominator

Using the zeros found in Step 1, we can factor the expression as a difference of squares:

$$2 - x^2 = (\sqrt{2} - x)(\sqrt{2} + x)$$

Step 3: Write the partial fractions with unknown constant numerators

We assign a constant to each of the distinct linear factors:

$$\frac{3}{(\sqrt{2} - x)(\sqrt{2} + x)} = \frac{A}{\sqrt{2} - x} + \frac{B}{\sqrt{2} + x}$$

Step 4: Sum the partial fractions

We combine the terms over a common denominator:

$$\frac{A(\sqrt{2} + x) + B(\sqrt{2} - x)}{(\sqrt{2} - x)(\sqrt{2} + x)}$$

Expanding the numerator and grouping the terms by the variable x :

$$\frac{A\sqrt{2} + Ax + B\sqrt{2} - Bx}{2 - x^2} = \frac{(A - B)x + \sqrt{2}(A + B)}{2 - x^2}$$

Step 5: Equal the numerators

We equate our new numerator to the original numerator, which is the constant 3. Since there is no x term in the original, its coefficient is 0:

$$(A - B)x + \sqrt{2}(A + B) = 0x + 3$$

This gives us the following system of equations:

$$\begin{cases} A - B = 0 \\ \sqrt{2}(A + B) = 3 \end{cases}$$

Step 6: Solve for the unknowns

From the first equation, $A = B$. Substituting this into the second equation:

$$\sqrt{2}(A + A) = 3 \Rightarrow 2A\sqrt{2} = 3$$

Solving for A :

$$A = \frac{3}{2\sqrt{2}}$$

Rationalizing the denominator by multiplying by $\frac{\sqrt{2}}{\sqrt{2}}$:

$$A = \frac{3\sqrt{2}}{4}$$

Since $A = B$, then $B = \frac{3\sqrt{2}}{4}$ as well.

Answer:
$$\frac{3\sqrt{2}}{4(\sqrt{2} - x)} + \frac{3\sqrt{2}}{4(\sqrt{2} + x)}$$

✔ **Solution 17: (d)**
$$\frac{1}{x - 0.01x^2}$$

Step 1: Find the zeros of the denominator

We set the quadratic denominator equal to zero to find the roots:

$$x - 0.01x^2 = 0$$

Factoring out the common term x :

$$x(1 - 0.01x) = 0$$

The zeros are $x = 0$ and $x = \frac{1}{0.01} = 100$.

Step 2: Factor the denominator

Using the common factor found in Step 1, the denominator is expressed as:

$$x - 0.01x^2 = x(1 - 0.01x)$$

Step 3: Write the partial fractions with unknown constant numerators

We assign a constant to each of the distinct linear factors:

$$\frac{1}{x(1 - 0.01x)} = \frac{A}{x} + \frac{B}{1 - 0.01x}$$

Step 4: Sum the partial fractions

We combine the terms by finding a common denominator:

$$\frac{A(1 - 0.01x) + Bx}{x(1 - 0.01x)}$$

Expanding the numerator and grouping the terms by the variable x :

$$\frac{A - 0.01Ax + Bx}{x - 0.01x^2} = \frac{(B - 0.01A)x + A}{x - 0.01x^2}$$

Step 5: Equal the numerators

We equate our new numerator to the original numerator, which is the constant 1. Since there is no x term in the original, its coefficient is 0:

$$(B - 0.01A)x + A = 0x + 1$$

This gives us the following system of equations:

$$\begin{cases} A = 1 \\ B - 0.01A = 0 \end{cases}$$

Step 6: Solve for the unknowns

From the first equation, we immediately have $A = 1$. Substituting this into the second equation:

$$B - 0.01(1) = 0 \Rightarrow B = 0.01$$

To simplify the second fraction $\frac{0.01}{1-0.01x}$, we multiply both its numerator and denominator by 100 to remove the decimals:

$$\frac{0.01 \cdot 100}{(1 - 0.01x) \cdot 100} = \frac{1}{100 - x}$$

Answer: $\frac{1}{x} + \frac{1}{100 - x}$

? Question 18

Compute the following indefinite integrals using the technique of partial fractions:

(a) $\int \frac{1}{x^2 - 9x + 20} dx$ (see the previous exercise)

(b) $\int \frac{dx}{x - x^2}$

(c) $\int \frac{0.01}{0.1x - 10x^2} dx$

(d) $\int \frac{dx}{x^2 - 12x + 36}$

💡 Tip: The 6-Step Program for Integration by Partial Fractions

To solve an integral using partial fractions, follow this complete systematic process:

Step 1: Find the zeros of the denominator by setting it equal to zero.

Step 2: Factor the denominator using the zeros (e.g., $(x - a)(x - b)$ or $(x - a)^2$).

Step 3: Write the setup with unknown constant numerators (A, B) over each linear factor.

Step 4: Sum the partial fractions by finding a common denominator to create a single fraction.

Step 5: Equal the numerators of your new expression to the original numerator to form a system of equations.

Step 6: Solve for the unknowns (A and B) by comparing coefficients or substituting values.

Finally, integrate the resulting simpler fractions, usually using the $\ln |f(x)|$ rule.

✓ **Solution 18: (a)** $\int \frac{1}{x^2 - 9x + 20} dx$

Step 1–6: Partial Fraction Decomposition

There is no need to perform these steps again, as we found the result in the previous exercise. Using our previous decomposition, we have:

$$\frac{1}{x^2 - 9x + 20} = \frac{1}{x - 5} - \frac{1}{x - 4}$$

Step 7: Integrate

Substitute these fractions into the integral and integrate term by term using the natural logarithm rule $\int \frac{1}{u} du = \ln |u| + C$:

$$\begin{aligned} \int \left(\frac{1}{x - 5} - \frac{1}{x - 4} \right) dx &= \int \frac{1}{x - 5} dx - \int \frac{1}{x - 4} dx \\ &= \ln |x - 5| - \ln |x - 4| + C \end{aligned}$$

To simplify, we apply the logarithm quotient property $\ln(a) - \ln(b) = \ln \left| \frac{a}{b} \right|$:

$$\ln \left| \frac{x - 5}{x - 4} \right| + C$$

Answer: $\ln \left| \frac{x - 5}{x - 4} \right| + C$

✓ **Solution 18: (b)** $\int \frac{1}{x - x^2} dx$

Step 1: Find the zeros of the denominator

We set the quadratic denominator equal to zero to find the roots:

$$x - x^2 = 0$$

Factoring out the common term x :

$$x(1 - x) = 0$$

The zeros are $x = 0$ and $x = 1$.

Step 2: Factor the denominator

Using the common factor found in Step 1, we rewrite the denominator in its linear factored form:

$$x - x^2 = x(1 - x)$$

Step 3: Write the partial fractions with unknown constant numerators

Since we have two distinct linear factors, we assign a constant to each:

$$\frac{1}{x(1 - x)} = \frac{A}{x} + \frac{B}{1 - x}$$

Step 4: Sum the partial fractions

To combine these, we find a common denominator by cross-multiplying the numerators:

$$\frac{A(1 - x) + Bx}{x(1 - x)}$$

Expanding the numerator and grouping the terms by the variable x :

$$\frac{A - Ax + Bx}{x - x^2} = \frac{(B - A)x + A}{x - x^2}$$

Step 5: Equal the numerators

We equate our new expression to the original numerator, 1. Since there is no x term in the original, its coefficient is 0:

$$(B - A)x + A = 0x + 1$$

This gives us the following system of equations:

$$\begin{cases} A = 1 \\ B - A = 0 \end{cases}$$

Step 6: Solve for the unknowns

From the first equation, we immediately have $A = 1$. Substituting this into the second equation:

$$B - 1 = 0 \Rightarrow B = 1$$

The resulting partial fraction decomposition is:

$$\frac{1}{x} + \frac{1}{1-x}$$

Step 7: Integrate

Substitute the decomposition into the integral and integrate term by term:

$$\int \left(\frac{1}{x} + \frac{1}{1-x} \right) dx = \int \frac{1}{x} dx + \int \frac{1}{1-x} dx$$

Applying the natural logarithm rule. Note that for the second term, we use the substitution $u = 1 - x$, which introduces a negative sign because $du = -dx$:

$$\ln |x| - \ln |1-x| + C$$

Applying the logarithm quotient property to combine the terms:

$$\ln \left| \frac{x}{1-x} \right| + C$$

Answer: $\ln \left| \frac{x}{1-x} \right| + C$

✓ **Solution 18: (c)** $\int \frac{0.01}{0.1x - 10x^2} dx$

Step 1: Find the zeros of the denominator

We set the quadratic denominator equal to zero to find the roots:

$$0.1x - 10x^2 = 0$$

Factoring out the common term $0.1x$:

$$0.1x(1 - 100x) = 0$$

The zeros are $x = 0$ and $x = \frac{1}{100} = 0.01$.

Step 2: Factor the denominator

Using the common factor found in Step 1, we rewrite the denominator in its linear factored form:

$$0.1x - 10x^2 = 10x(0.01 - x)$$

Step 3: Write the partial fractions with unknown constant numerators

First, we simplify the integrand by dividing the numerator and denominator by 10:

$$\frac{0.01}{10x(0.01 - x)} = \frac{0.001}{x(0.01 - x)}$$

Now, we assign a constant to each distinct linear factor:

$$\frac{0.001}{x(0.01 - x)} = \frac{A}{x} + \frac{B}{0.01 - x}$$

Step 4: Sum the partial fractions

To combine these, we find a common denominator:

$$\frac{A(0.01 - x) + Bx}{x(0.01 - x)}$$

Expanding the numerator and grouping the terms by the variable x :

$$\frac{0.01A - Ax + Bx}{0.01x - x^2} = \frac{(B - A)x + 0.01A}{0.01x - x^2}$$

Step 5: Equal the numerators

We equate our new numerator to the simplified original numerator, 0.001. Since there is no x term, its coefficient is 0:

$$(B - A)x + 0.01A = 0x + 0.001$$

This gives us the following system of equations:

$$\begin{cases} 0.01A = 0.001 \\ B - A = 0 \end{cases}$$

Step 6: Solve for the unknowns

From the first equation, $A = \frac{0.001}{0.01} = 0.1$. Substituting this into the second equation:

$$B - 0.1 = 0 \Rightarrow B = 0.1$$

The resulting partial fraction decomposition is:

$$\frac{0.1}{x} + \frac{0.1}{0.01 - x}$$

Step 7: Integrate

Substitute the decomposition into the integral and integrate term by term:

$$\int \left(\frac{0.1}{x} + \frac{0.1}{0.01 - x} \right) dx = 0.1 \int \frac{1}{x} dx + 0.1 \int \frac{1}{0.01 - x} dx$$

Applying the natural logarithm rule. Note the negative sign for the second term due to the derivative of the denominator being -1 :

$$0.1 \ln |x| - 0.1 \ln |0.01 - x| + C$$

Factoring out 0.1 and applying the logarithm quotient property:

$$0.1 \ln \left| \frac{x}{0.01 - x} \right| + C$$

Answer: $0.1 \ln \left| \frac{x}{0.01 - x} \right| + C$

✓ **Solution 18: (d)** $\int \frac{1}{x^2 - 12x + 36} dx$

Step 1: Find the zeros of the denominator

We set the quadratic denominator equal to zero to find the roots:

$$x^2 - 12x + 36 = 0$$

Recognizing this as a perfect square trinomial:

$$(x - 6)^2 = 0$$

The zero is $x = 6$ with a multiplicity of 2.

Step 2: Factor the denominator

Based on the zero found in Step 1, the denominator factors into a repeated linear factor:

$$x^2 - 12x + 36 = (x - 6)^2$$

Step 3: Write the partial fractions with unknown constant numerators

For a repeated linear factor $(x - a)^2$, the decomposition setup requires two terms:

$$\frac{1}{(x - 6)^2} = \frac{A}{x - 6} + \frac{B}{(x - 6)^2}$$

Step 4: Sum the partial fractions

We combine the terms by finding the least common denominator, which is $(x - 6)^2$:

$$\frac{A(x - 6) + B}{(x - 6)^2}$$

Expanding the numerator and grouping the terms by the variable x :

$$\frac{Ax + (B - 6A)}{(x - 6)^2}$$

Step 5: Equal the numerators

We equate our new numerator to the original numerator, 1. Since there is no x term, its coefficient is 0:

$$Ax + (B - 6A) = 0x + 1$$

This gives us the following system of equations:

$$\begin{cases} A = 0 \\ B - 6A = 1 \end{cases}$$

Step 6: Solve for the unknowns

From the first equation, we have $A = 0$. Substituting this into the second equation:

$$B - 6(0) = 1 \Rightarrow B = 1$$

This confirms the fraction was already in its simplest form for integration: $\frac{1}{(x-6)^2}$.

Step 7: Integrate

We rewrite the expression using a negative exponent and apply the power rule $\int u^n du = \frac{u^{n+1}}{n+1} + C$:

$$\int (x-6)^{-2} dx = \frac{(x-6)^{-2+1}}{-2+1} + C = \frac{(x-6)^{-1}}{-1} + C$$

Simplifying the final result:

$$-\frac{1}{x-6} + C$$

Answer: $-\frac{1}{x-6} + C$

Booklet 1 - Indefinite Integrals

? Question 1

Determine all antiderivatives $F(x)$ of $f(x) = 3^x + e^{3x}$ that also satisfy $F(0) = \frac{1}{\ln 3} + \frac{1}{3}$.

✓ Solution 1: Determine the antiderivative $F(x)$

To find the specific antiderivative, we first perform the indefinite integration of $f(x) = 3^x + e^{3x}$:

$$F(x) = \int (3^x + e^{3x}) dx$$

Step 1: Integrate the exponential terms

We apply the rule for general exponential bases $\int a^x dx = \frac{a^x}{\ln a}$ and the rule for e^{kx} which is $\int e^{kx} dx = \frac{1}{k}e^{kx}$:

$$F(x) = \frac{3^x}{\ln 3} + \frac{e^{3x}}{3} + C$$

Step 2: Solve for the constant C

We use the given initial condition $F(0) = \frac{1}{\ln 3} + \frac{1}{3}$:

$$\frac{1}{\ln 3} + \frac{1}{3} = \frac{3^0}{\ln 3} + \frac{e^{3(0)}}{3} + C$$

$$\frac{1}{\ln 3} + \frac{1}{3} = \frac{1}{\ln 3} + \frac{1}{3} + C$$

Subtracting the identical terms from both sides:

$$0 = C$$

Answer: $F(x) = \frac{3^x}{\ln 3} + \frac{e^{3x}}{3}$

Question 2

Compute the following indefinite integrals using substitution:

(a) $\int (3^x + 3^{-x}) dx$

(g) $\int \frac{\sqrt{\ln x}}{x} dx$

(b) $\int (e^x + e^{-x} + x^e + 2^{-e}) dx$

(h) $\int \frac{x}{x^2 + 3} \ln(x^2 + 3) dx$

(c) $\int 2x\sqrt{x^2 + 1} dx$

(i) $\int \frac{(\ln(3 - x))^4}{3 - x} dx$

(d) $\int \frac{2x^3}{x^4 - 1} dx$

(j) $\int \frac{e^{3x}}{e^{3x} + 2} dx$

(e) $\int \frac{\ln(1 - x)}{x - 1} dx$

(k) $\int (e^{5-2x})^2 dx$

(f) $\int 3(x - 1)e^{x^2-2x} dx$

(l) $\int \frac{ye^{y^2}}{e^{y^2} - 4} dy$

✓ Solution 2: (a) $\int (3^x + 3^{-x}) dx$

We split the integral into two parts. For the second term, we use a simple substitution $u = -x$.

$$\int 3^x dx + \int 3^{-x} dx$$

For the second part:

$$u = -x$$

$$du = -1 dx \Rightarrow dx = -du$$

Substituting and using the exponential rule $\int a^u du = \frac{a^u}{\ln a}$:

$$\frac{3^x}{\ln 3} + \int 3^u (-du) = \frac{3^x}{\ln 3} - \frac{3^u}{\ln 3} + C$$

Finally, we replace u with $-x$ to return to the original variable x .

Answer: $\frac{3^x - 3^{-x}}{\ln 3} + C$

✔ **Solution 2: (b)** $\int (e^x + e^{-x} + x^e + 2^{-e}) dx$

We integrate each term separately. Note that x^e follows the power rule (base is variable) and 2^{-e} is a constant.

$$\int e^x dx + \int e^{-x} dx + \int x^e dx + \int 2^{-e} dx$$

For the second term, we use $u = -x$, $du = -dx$:

$$e^x - e^{-x} + \frac{x^{e+1}}{e+1} + 2^{-e}x + C$$

The integral of e^{-x} becomes $-e^{-x}$ because of the chain rule/substitution. The constant 2^{-e} simply gains an x factor.

Answer: $e^x - e^{-x} + \frac{x^{e+1}}{e+1} + 2^{-e}x + C$

✔ **Solution 2: (c)** $\int 2x\sqrt{x^2 + 1} dx$

We choose $u = x^2 + 1$ because its derivative $2x$ is present exactly as a multiplier.

$$u = x^2 + 1$$

$$du = 2x dx$$

Substituting these into the integral and rewriting the root as a power:

$$\int \sqrt{u} \, du = \int u^{1/2} \, du = \frac{u^{3/2}}{3/2} + C = \frac{2}{3} u^{3/2} + C$$

Finally, we replace u with $x^2 + 1$ to return to the original variable x .

Answer: $\frac{2}{3}(x^2 + 1)\sqrt{x^2 + 1} + C$

✓ **Solution 2: (d)** $\int \frac{2x^3}{x^4 - 1} \, dx$

We choose $u = x^4 - 1$ because its derivative $4x^3$ matches the power of the numerator.

$$u = x^4 - 1$$

$$du = 4x^3 \, dx \quad \Rightarrow \quad 2x^3 \, dx = \frac{du}{2}$$

Substituting into the integral:

$$\int \frac{1}{u} \cdot \frac{du}{2} = \frac{1}{2} \int \frac{1}{u} \, du = \frac{1}{2} \ln |u| + C$$

Finally, we replace u with $x^4 - 1$ to return to the original variable x .

Answer: $\frac{1}{2} \ln |x^4 - 1| + C$

✓ **Solution 2: (e)** $\int \frac{\ln(1-x)}{x-1} dx$

We choose $u = \ln(1-x)$ because its derivative involves $1/(1-x)$. Note that $x-1 = -(1-x)$.

$$u = \ln(1-x)$$

$$du = \frac{1}{1-x} \cdot (-1) dx = \frac{1}{x-1} dx$$

Substituting u and du directly into the integral:

$$\int u du = \frac{u^2}{2} + C$$

Finally, we replace u with $\ln(1-x)$ to return to the original variable x .

Answer: $\frac{1}{2}(\ln(1-x))^2 + C$

✓ **Solution 2: (f)** $\int 3(x-1)e^{x^2-2x} dx$

We choose the exponent $u = x^2 - 2x$ because its derivative $2x - 2 = 2(x-1)$ is proportional to the term outside.

$$u = x^2 - 2x$$

$$du = (2x - 2) dx = 2(x-1) dx \Rightarrow (x-1) dx = \frac{du}{2}$$

Substituting into the integral and moving the constant 3 outside:

$$3 \int e^u \cdot \frac{du}{2} = \frac{3}{2} \int e^u du = \frac{3}{2} e^u + C$$

Finally, we replace u with $x^2 - 2x$ to return to the original variable x .

Answer: $\frac{3}{2}e^{x^2-2x} + C$

✓ **Solution 2: (g)** $\int \frac{\sqrt{\ln x}}{x} dx$

We choose $u = \ln x$ because its derivative $1/x$ is present in the integrand.

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

Substituting into the integral and rewriting the root as a power:

$$\int \sqrt{u} du = \int u^{1/2} du = \frac{u^{3/2}}{3/2} + C = \frac{2}{3}u^{3/2} + C$$

Finally, we replace u with $\ln x$ to return to the original variable x .

Answer: $\frac{2}{3} \ln x \sqrt{\ln x} + C$

✓ **Solution 2: (h)** $\int \frac{x}{x^2 + 3} \ln(x^2 + 3) dx$

We choose $u = \ln(x^2 + 3)$ because its derivative involves $x/(x^2 + 3)$.

$$u = \ln(x^2 + 3)$$

$$du = \frac{2x}{x^2 + 3} dx \Rightarrow \frac{x}{x^2 + 3} dx = \frac{du}{2}$$

Substituting u and du into the integral:

$$\int u \cdot \frac{du}{2} = \frac{1}{2} \left(\frac{u^2}{2} \right) + C = \frac{1}{4} u^2 + C$$

Finally, we replace u with $\ln(x^2 + 3)$ to return to the original variable x .

Answer: $\frac{1}{4}(\ln(x^2 + 3))^2 + C$

✓ **Solution 2: (i)** $\int \frac{(\ln(3 - x))^4}{3 - x} dx$

We choose $u = \ln(3 - x)$. Its derivative involves $1/(3 - x)$, but we must account for the chain rule.

$$u = \ln(3 - x)$$

$$du = \frac{1}{3 - x} \cdot (-1) dx \Rightarrow \frac{1}{3 - x} dx = -du$$

Substituting these into the integral:

$$\int u^4 (-du) = - \int u^4 du = -\frac{u^5}{5} + C$$

Finally, we replace u with $\ln(3 - x)$ to return to the original variable x .

Answer: $-\frac{1}{5}(\ln(3 - x))^5 + C$

✓ **Solution 2: (j)** $\int \frac{e^{3x}}{e^{3x} + 2} dx$

We choose the denominator $u = e^{3x} + 2$ because its derivative $3e^{3x}$ matches the numerator.

$$u = e^{3x} + 2$$

$$du = 3e^{3x} dx \Rightarrow e^{3x} dx = \frac{du}{3}$$

Substituting into the integral:

$$\int \frac{1}{u} \cdot \frac{du}{3} = \frac{1}{3} \ln |u| + C$$

Finally, we replace u with $e^{3x} + 2$ to return to the original variable x .

Answer: $\frac{1}{3} \ln(e^{3x} + 2) + C$

✓ **Solution 2: (k)** $\int (e^{5-2x})^2 dx$

First, we simplify the integrand using the property $(e^a)^b = e^{ab}$:

$$\int e^{2(5-2x)} dx = \int e^{10-4x} dx$$

Now, we choose $u = 10 - 4x$:

$$u = 10 - 4x$$

$$du = -4 dx \Rightarrow dx = -\frac{du}{4}$$

Substituting and integrating:

$$\int e^u \cdot \left(-\frac{du}{4}\right) = -\frac{1}{4} e^u + C$$

Finally, we replace u with $10 - 4x$ to return to the original variable x .

Answer: $-\frac{1}{4}e^{10-4x} + C$

✓ **Solution 2: (I)** $\int \frac{ye^{y^2}}{e^{y^2} - 4} dy$

We choose $u = e^{y^2} - 4$ because its derivative involves ye^{y^2} due to the chain rule.

$$u = e^{y^2} - 4$$

$$du = e^{y^2} \cdot 2y dy \Rightarrow ye^{y^2} dy = \frac{du}{2}$$

Substituting into the integral:

$$\int \frac{1}{u} \cdot \frac{du}{2} = \frac{1}{2} \ln |u| + C$$

Finally, we replace u with $e^{y^2} - 4$ to return to the original variable y .

Answer: $\frac{1}{2} \ln |e^{y^2} - 4| + C$

Question 3

Using integration by parts, compute the following indefinite integrals:

(a) $\int x \cdot (2x + 1)^4 dx$

(b) $\int x \cdot \ln(1 + 4x^2) dx$

(c) $\int \frac{x^3}{\sqrt{1 + x^2}} dx$

Hints:

- For (b): first use the substitution $u = 1 + 4x^2$.
- For (c): $x^3 = x^2 \cdot x$.

✓ Solution 3: (a) $\int x \cdot (2x + 1)^4 dx$

The integrand contains two Algebraic terms. We choose $f(x) = x$ to differentiate because its derivative is 1, which will simplify the remaining integral.

$$f(x) = x \Rightarrow f'(x) = 1$$

$$g'(x) = (2x + 1)^4 \Rightarrow g(x) = \frac{(2x + 1)^5}{5}$$

Applying the integration by parts formula $\int f g' = f g - \int f' g$:

$$\int x(2x + 1)^4 dx = \frac{x(2x + 1)^5}{5} - \int \frac{(2x + 1)^5}{5} dx$$

Performing the final integration:

$$\frac{x(2x + 1)^5}{5} - \frac{1}{5} \left(\frac{(2x + 1)^6}{6} \right) + C$$

Answer: $\frac{1}{10}x(2x+1)^5 - \frac{1}{120}(2x+1)^6 + C$

✓ **Solution 3: (b)** $\int x \cdot \ln(1 + 4x^2) dx$

Following the hint, we first use substitution $u = 1 + 4x^2$, so $du = 8x dx$, meaning $x dx = \frac{du}{8}$:

$$\int \ln(u) \cdot \frac{du}{8} = \frac{1}{8} \int \ln u du$$

Now we apply integration by parts to $\int \ln u du$ where $f(u) = \ln u$ and $g'(u) = 1$:

$$f(u) = \ln u \Rightarrow f'(u) = \frac{1}{u}$$

$$g'(u) = 1 \Rightarrow g(u) = u$$

$$\frac{1}{8} \left(u \ln u - \int 1 du \right) = \frac{1}{8} (u \ln u - u) + C$$

Finally, we substitute back $u = 1 + 4x^2$:

Answer: $\frac{1}{8}(1 + 4x^2) \ln(1 + 4x^2) - \frac{1}{8}(1 + 4x^2) + C$

✓ **Solution 3: (c)** $\int \frac{x^3}{\sqrt{1+x^2}} dx$

Following the hint, we rewrite the numerator as $x^2 \cdot x$. We choose $f(x) = x^2$ to differentiate and $g'(x) = \frac{x}{\sqrt{1+x^2}}$ to integrate.

To find $g(x)$, we perform a formal substitution for the side integral $\int \frac{x}{\sqrt{1+x^2}} dx$:

$$w = 1 + x^2$$

$$dw = 2x dx \Rightarrow x dx = \frac{dw}{2}$$

$$g(x) = \int \frac{1}{\sqrt{w}} \cdot \frac{dw}{2} = \frac{1}{2} \int w^{-1/2} dw = \frac{1}{2}(2w^{1/2}) = \sqrt{1+x^2}$$

Now we organize our terms for Integration by Parts:

$$f(x) = x^2 \Rightarrow f'(x) = 2x$$

$$g'(x) = \frac{x}{\sqrt{1+x^2}} \Rightarrow g(x) = \sqrt{1+x^2}$$

Applying the formula $\int f g' = f g - \int f' g$:

$$\int x^2 \cdot \frac{x}{\sqrt{1+x^2}} dx = x^2 \sqrt{1+x^2} - \int 2x \sqrt{1+x^2} dx$$

To solve the remaining integral $\int 2x \sqrt{1+x^2} dx$, we use substitution again with $v = 1 + x^2$ and $dv = 2x dx$:

$$\int \sqrt{v} dv = \int v^{1/2} dv = \frac{v^{3/2}}{3/2} = \frac{2}{3}(1+x^2)^{3/2}$$

Combining everything together:

$$x^2 \sqrt{1+x^2} - \frac{2}{3}(1+x^2)^{3/2} + C$$

Answer: $x^2 \sqrt{1+x^2} - \frac{2}{3}(1+x^2) \sqrt{1+x^2} + C$

Question 4

4. Using partial fractions, compute the following indefinite integrals:

(a) $\int \frac{x+3}{x^2+3x+2} dx$

(b) $\int \frac{1}{-x^2+x+12} dx$

(c) $\int \frac{1}{x^2+4x} dx$

✓ **Solution 4: (a)** $\int \frac{x+3}{x^2+3x+2} dx$

Step 1: Find the zeros of the denominator

We set the quadratic denominator equal to zero to find the roots:

$$x^2 + 3x + 2 = 0$$

Factoring the quadratic:

$$(x+1)(x+2) = 0$$

The zeros are $x = -1$ and $x = -2$.

Step 2: Factor the denominator

Using the factors found in Step 1, we rewrite the denominator in its linear factored form:

$$x^2 + 3x + 2 = (x+1)(x+2)$$

Step 3: Write the partial fractions with unknown constant numerators

We assign a constant to each distinct linear factor:

$$\frac{x+3}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

Step 4: Sum the partial fractions

To combine these, we find a common denominator:

$$\frac{A(x+2) + B(x+1)}{(x+1)(x+2)}$$

Expanding the numerator and grouping the terms by the variable x :

$$\frac{Ax + 2A + Bx + B}{(x+1)(x+2)} = \frac{(A+B)x + (2A+B)}{(x+1)(x+2)}$$

Step 5: Equal the numerators

We equate our new numerator to the original numerator, $x + 3$:

$$(A+B)x + (2A+B) = 1x + 3$$

This gives us the following system of equations:

$$\begin{cases} A + B = 1 \\ 2A + B = 3 \end{cases}$$

Step 6: Solve for the unknowns

Subtracting the first equation from the second:

$$(2A+B) - (A+B) = 3-1 \Rightarrow A = 2$$

Substituting $A = 2$ into the first equation:

$$2 + B = 1 \Rightarrow B = -1$$

The resulting partial fraction decomposition is:

$$\frac{2}{x+1} - \frac{1}{x+2}$$

Step 7: Integrate

Substitute the decomposition into the integral and integrate term by term:

$$\int \left(\frac{2}{x+1} - \frac{1}{x+2} \right) dx = 2 \int \frac{1}{x+1} dx - \int \frac{1}{x+2} dx$$

Applying the natural logarithm rule:

$$2 \ln |x + 1| - \ln |x + 2| + C$$

Using the logarithm power and quotient properties:

$$\ln(x + 1)^2 - \ln |x + 2| + C = \ln \frac{(x + 1)^2}{|x + 2|} + C$$

Answer: $\ln \frac{(x + 1)^2}{|x + 2|} + C$

✓ **Solution 4: (b)** $\int \frac{1}{-x^2 + x + 12} dx$

Step 1: Find the zeros of the denominator

We set the quadratic denominator equal to zero:

$$-x^2 + x + 12 = 0 \Rightarrow x^2 - x - 12 = 0$$

Factoring:

$$(x - 4)(x + 3) = 0$$

The zeros are $x = 4$ and $x = -3$.

Step 2: Factor the denominator

We rewrite the denominator in its linear factored form, accounting for the leading negative sign:

$$-x^2 + x + 12 = -(x - 4)(x + 3) = (4 - x)(x + 3)$$

Step 3: Write the partial fractions with unknown constant numerators

Assign a constant to each distinct linear factor:

$$\frac{1}{(4 - x)(x + 3)} = \frac{A}{4 - x} + \frac{B}{x + 3}$$

Step 4: Sum the partial fractions

Find a common denominator:

$$\frac{A(x+3) + B(4-x)}{(4-x)(x+3)} = \frac{(A-B)x + (3A+4B)}{-x^2 + x + 12}$$

Step 5: Equal the numerators

Equate the new numerator to the original numerator, 1:

$$(A-B)x + (3A+4B) = 0x + 1$$

This gives us the system:

$$\begin{cases} A - B = 0 & \Rightarrow A = B \\ 3A + 4B = 1 \end{cases}$$

Step 6: Solve for the unknowns

Substituting $A = B$ into the second equation:

$$3B + 4B = 1 \Rightarrow 7B = 1 \Rightarrow B = \frac{1}{7}, A = \frac{1}{7}$$

The decomposition is:

$$\frac{1/7}{4-x} + \frac{1/7}{x+3}$$

Step 7: Integrate

Substitute and integrate. Note the negative sign for the first term due to the derivative of $(4-x)$ being -1 :

$$\frac{1}{7} \int \frac{1}{4-x} dx + \frac{1}{7} \int \frac{1}{x+3} dx = -\frac{1}{7} \ln |4-x| + \frac{1}{7} \ln |x+3| + C$$

Using the logarithm quotient property:

$$\frac{1}{7} \ln \left| \frac{x+3}{4-x} \right| + C = \frac{1}{7} \ln \left| \frac{x+3}{x-4} \right| + C$$

Answer: $\frac{1}{7} \ln \left| \frac{x+3}{x-4} \right| + C$

✓ **Solution 4: (c)** $\int \frac{1}{x^2 + 4x} dx$

Step 1: Find the zeros of the denominator

Set the denominator to zero:

$$x^2 + 4x = 0$$

Factoring out x :

$$x(x + 4) = 0$$

The zeros are $x = 0$ and $x = -4$.

Step 2: Factor the denominator

The factored form is:

$$x^2 + 4x = x(x + 4)$$

Step 3: Write the partial fractions with unknown constant numerators

Assign constants to the linear factors:

$$\frac{1}{x(x + 4)} = \frac{A}{x} + \frac{B}{x + 4}$$

Step 4: Sum the partial fractions

Find a common denominator:

$$\frac{A(x + 4) + Bx}{x(x + 4)} = \frac{(A + B)x + 4A}{x^2 + 4x}$$

Step 5: Equal the numerators

Equate the new numerator to the original numerator, 1:

$$(A + B)x + 4A = 0x + 1$$

This gives the system:

$$\begin{cases} 4A = 1 & \Rightarrow & A = 1/4 \\ A + B = 0 \end{cases}$$

Step 6: Solve for the unknowns

Substituting $A = 1/4$ into the second equation:

$$1/4 + B = 0 \Rightarrow B = -1/4$$

The decomposition is:

$$\frac{1/4}{x} - \frac{1/4}{x+4}$$

Step 7: Integrate

Substitute and integrate term by term:

$$\frac{1}{4} \int \frac{1}{x} dx - \frac{1}{4} \int \frac{1}{x+4} dx = \frac{1}{4} \ln |x| - \frac{1}{4} \ln |x+4| + C$$

Using the logarithm quotient property:

$$\frac{1}{4} \ln \left| \frac{x}{x+4} \right| + C$$

Answer: $\frac{1}{4} \ln \left| \frac{x}{x+4} \right| + C$

Question 5

Compute the following indefinite integrals:

(a) $\int s \cdot e^{s^2} ds$

(f) $\int e^{ax+b} dx \quad (a \neq 0)$

(b) $\int v \ln v dv$

(g) $\int \frac{1}{x^2 + 4x + 4} dx$

(c) $\int x \cdot (3e^x + 2)^2 dx$

(h) $\int \frac{x \ln(x^2 + 1)}{x^2 + 1} dx$

(d) $\int \frac{x}{\sqrt{1-x^2}} dx$

(i) $\int x \cdot \ln(x^2 - 1) dx$

(e) $\int \frac{1}{x^2 + 4x + 3} dx$

(j) $\int \frac{x+2}{(x^2 + 4x + 9)^3} dx$

✓ Solution 5: (a) $\int s \cdot e^{s^2} ds$

Step 1: Choose a substitution

Let $u = s^2$. Then, its derivative is:

$$du = 2s ds \quad \Rightarrow \quad s ds = \frac{1}{2} du$$

Step 2: Substitute and integrate

Substitute these into the integral:

$$\int e^u \cdot \frac{1}{2} du = \frac{1}{2} \int e^u du$$

Integrating e^u gives $e^u + C$:

$$\frac{1}{2} e^u + C$$

Step 3: Back-substitute

Replace u with the original expression s^2 :

$$\frac{1}{2}e^{s^2} + C$$

Answer: $\frac{1}{2}e^{s^2} + C$

✓ Solution 5: (b) $\int v \ln v \, dv$

Step 1: Choose terms for Integration by Parts

Using the LAE rule, we choose the logarithmic term to differentiate:

$$f(v) = \ln v \Rightarrow f'(v) = \frac{1}{v}$$

$$g'(v) = v \Rightarrow g(v) = \frac{1}{2}v^2$$

Step 2: Apply the Integration by Parts formula

Using $\int f g' = f g - \int f' g$:

$$\frac{1}{2}v^2 \ln v - \int \left(\frac{1}{v} \cdot \frac{1}{2}v^2 \right) dv$$

Simplify the integrand:

$$\frac{1}{2}v^2 \ln v - \frac{1}{2} \int v \, dv$$

Step 3: Final Integration

Integrate the remaining v term:

$$\frac{1}{2}v^2 \ln v - \frac{1}{2} \left(\frac{1}{2}v^2 \right) + C = \frac{1}{2}v^2 \ln v - \frac{1}{4}v^2 + C$$

Answer: $\frac{1}{2}v^2 \ln v - \frac{1}{4}v^2 + C$

✓ Solution 5: (c) $\int x \cdot (3e^x + 2)^2 dx$

We start by squaring the term in the parentheses. This turns the problem into a sum of simpler parts that we can integrate one by one.

Step 1: Expand the squared term

Multiply out the $(3e^x + 2)^2$ and then distribute the x :

$$(3e^x + 2)(3e^x + 2) = 9e^{2x} + 12e^x + 4$$

$$\int x(9e^{2x} + 12e^x + 4) dx = \int (9xe^{2x} + 12xe^x + 4x) dx$$

Step 2: Integrate term by term

We use Integration by Parts for the first two terms. Following the LAE order, we pick x to differentiate and the e terms to integrate:

- For $9 \int xe^{2x} dx$:

$$f(x) = x \Rightarrow f'(x) = 1$$

$$g'(x) = e^{2x} \Rightarrow g(x) = \frac{1}{2}e^{2x}$$

$$\text{Result: } 9 \left(\frac{1}{2}xe^{2x} - \int \frac{1}{2}e^{2x} dx \right) = \frac{9}{2}xe^{2x} - \frac{9}{4}e^{2x}$$

- For $12 \int xe^x dx$:

$$f(x) = x \Rightarrow f'(x) = 1$$

$$g'(x) = e^x \Rightarrow g(x) = e^x$$

$$\text{Result: } 12(xe^x - \int e^x dx) = 12xe^x - 12e^x$$

- For the last part:

$$\int 4x dx = 2x^2$$

Step 3: Combine results

We put all the pieces back together:

$$\begin{aligned} \frac{9}{2}xe^{2x} - \frac{9}{4}e^{2x} + 12xe^x - 12e^x + 2x^2 + C \\ \left(\frac{9}{2}x - \frac{9}{4}\right)e^{2x} + 12(x-1)e^x + 2x^2 + C \end{aligned}$$

Answer: $\left(\frac{9}{2}x - \frac{9}{4}\right)e^{2x} + 12(x-1)e^x + 2x^2 + C$

✓ **Solution 5: (d)** $\int \frac{x}{\sqrt{1-x^2}} dx$

The numerator is almost the derivative of the expression inside the square root. Because of this, we can use substitution to turn the integral into a simple power rule problem.

Step 1: Choose a substitution

We let u be the expression under the radical:

$$u = 1 - x^2$$

Now we find the derivative to replace dx :

$$du = -2x dx \Rightarrow x dx = -\frac{1}{2}du$$

Step 2: Substitute and integrate

We replace the terms in the integral with our u and du :

$$\int \frac{1}{\sqrt{u}} \left(-\frac{1}{2}\right) du = -\frac{1}{2} \int u^{-1/2} du$$

Using the power rule (add 1 to the exponent and divide):

$$-\frac{1}{2} \left(\frac{u^{1/2}}{1/2} \right) + C = -u^{1/2} + C$$

Step 3: Back-substitute

Finally, we swap u back for our original x expression:

$$-\sqrt{1-x^2} + C$$

Answer: $-\sqrt{1-x^2} + C$

✓ **Solution 5: (e)** $\int \frac{1}{x^2 + 4x + 3} dx$

The denominator is a quadratic that can be broken down into two simpler parts. We use partial fractions to split the single fraction into two smaller ones that are easier to integrate.

Step 1: Find the zeros of the denominator

We set the bottom equal to zero to find where it breaks:

$$x^2 + 4x + 3 = 0$$

Factoring the quadratic gives:

$$(x + 1)(x + 3) = 0$$

The zeros are $x = -1$ and $x = -3$.

Step 2: Factor the denominator

We rewrite the integral using these factors:

$$x^2 + 4x + 3 = (x + 1)(x + 3)$$

Step 3: Write the partial fractions with unknown constants

We split the fraction and assign A and B to the numerators:

$$\frac{1}{(x + 1)(x + 3)} = \frac{A}{x + 1} + \frac{B}{x + 3}$$

Step 4: Sum the partial fractions

We combine them back over a common denominator:

$$\frac{A(x+3) + B(x+1)}{(x+1)(x+3)}$$

Grouping the x terms together:

$$\frac{(A+B)x + (3A+B)}{x^2 + 4x + 3}$$

Step 5: Equal the numerators

We compare this to our original numerator, which is just 1 (or $0x + 1$):

$$\begin{cases} A + B = 0 & \Rightarrow B = -A \\ 3A + B = 1 \end{cases}$$

Step 6: Solve for the unknowns

Substituting $B = -A$ into the second equation:

$$3A - A = 1 \Rightarrow 2A = 1 \Rightarrow A = 1/2$$

Since $B = -A$, then $B = -1/2$. The new fractions are:

$$\frac{1/2}{x+1} - \frac{1/2}{x+3}$$

Step 7: Integrate

Now we integrate each part separately:

$$\frac{1}{2} \int \frac{1}{x+1} dx - \frac{1}{2} \int \frac{1}{x+3} dx$$

This gives us natural logs:

$$\frac{1}{2} \ln |x+1| - \frac{1}{2} \ln |x+3| + C$$

Using log rules to combine them:

$$\frac{1}{2} \ln \left| \frac{x+1}{x+3} \right| + C$$

Answer: $\frac{1}{2} \ln \left| \frac{x+1}{x+3} \right| + C$

✓ Solution 5: (f) $\int e^{ax+b} dx \quad (a \neq 0)$

This is a linear substitution problem. Since the exponent is a first-degree polynomial ($ax + b$), we can use a simple u -substitution to turn it into a standard exponential integral.

Step 1: Choose a substitution

We let u be the entire exponent:

$$u = ax + b$$

Now we find the derivative to replace dx :

$$du = a dx \quad \Rightarrow \quad dx = \frac{1}{a} du$$

Step 2: Substitute and integrate

We replace the terms in the integral. Since a is a constant, we move it outside:

$$\int e^u \cdot \frac{1}{a} du = \frac{1}{a} \int e^u du$$

The integral of e^u is just e^u :

$$\frac{1}{a} e^u + C$$

Step 3: Back-substitute

Finally, we replace u with the original $ax + b$ expression:

$$\frac{1}{a} e^{ax+b} + C$$

Answer: $\frac{1}{a} e^{ax+b} + C$

✓ **Solution 5: (g)** $\int \frac{1}{x^2 + 4x + 4} dx$

While this looks like a partial fractions problem, the denominator is actually a "perfect square." This means we can rewrite it as a single power and use the power rule instead of splitting it into multiple fractions.

Step 1: Find the zeros of the denominator

We set the quadratic equal to zero:

$$x^2 + 4x + 4 = 0$$

Factoring the quadratic gives:

$$(x + 2)(x + 2) = (x + 2)^2 = 0$$

The only zero is $x = -2$.

Step 2: Factor the denominator

We rewrite the integral using this squared form:

$$\int \frac{1}{(x + 2)^2} dx = \int (x + 2)^{-2} dx$$

Step 3: Choose a substitution

Let u be the inside of the parentheses:

$$u = x + 2 \Rightarrow du = dx$$

Step 4: Substitute and integrate

Now we apply the power rule (add 1 to the exponent and divide):

$$\int u^{-2} du = \frac{u^{-1}}{-1} + C = -\frac{1}{u} + C$$

Step 5: Back-substitute

Swap u back for our $x + 2$ expression:

$$-\frac{1}{x + 2} + C$$

Answer: $\frac{-1}{x+2} + C$

✓ **Solution 5: (h)** $\int \frac{x \ln(x^2 + 1)}{x^2 + 1} dx$

This problem looks complicated, but notice that the derivative of $x^2 + 1$ is $2x$, which is similar to the x we see in the numerator. Also, the derivative of $\ln(x^2 + 1)$ is $\frac{2x}{x^2+1}$. This suggests we can use a substitution to simplify the whole fraction at once.

Step 1: Choose a substitution

We let u be the natural log term:

$$u = \ln(x^2 + 1)$$

Now we find the derivative to replace the other parts of the integral:

$$du = \frac{1}{x^2 + 1} \cdot 2x dx \Rightarrow \frac{x}{x^2 + 1} dx = \frac{1}{2} du$$

Step 2: Substitute and integrate

We replace the log with u and the remaining fraction with $\frac{1}{2} du$:

$$\int u \cdot \frac{1}{2} du = \frac{1}{2} \int u^1 du$$

Using the power rule:

$$\frac{1}{2} \left(\frac{u^2}{2} \right) + C = \frac{1}{4} u^2 + C$$

Step 3: Back-substitute

Swap u back for the original x expression:

$$\frac{1}{4} (\ln(x^2 + 1))^2 + C$$

Answer: $\frac{1}{4}(\ln(x^2 + 1))^2 + C$

✓ **Solution 5: (i)** $\int x \cdot \ln(x^2 - 1) dx$

We have an algebraic part (x) and a logarithmic part ($\ln(x^2 - 1)$). While we could use Integration by Parts right away, it is much easier to first use a substitution to simplify the inside of the log.

Step 1: Choose a substitution

Let w be the expression inside the log:

$$w = x^2 - 1 \Rightarrow dw = 2x dx \Rightarrow x dx = \frac{1}{2}dw$$

The integral becomes:

$$\frac{1}{2} \int \ln(w) dw$$

Step 2: Integrate by Parts

Now we integrate $\ln(w)$ by parts. Following the LAE order, we choose the log to differentiate:

$$f(w) = \ln(w) \Rightarrow f'(w) = \frac{1}{w}$$

$$g'(w) = 1 \Rightarrow g(w) = w$$

Applying the method:

$$\begin{aligned} \frac{1}{2} \left(w \ln(w) - \int w \cdot \frac{1}{w} dw \right) &= \frac{1}{2} \left(w \ln(w) - \int 1 dw \right) \\ &= \frac{1}{2} w \ln(w) - \frac{1}{2} w + C \end{aligned}$$

Step 3: Back-substitute

Replace w with $x^2 - 1$:

$$\frac{1}{2}(x^2 - 1) \ln(x^2 - 1) - \frac{1}{2}(x^2 - 1) + C$$

Answer: $\frac{1}{2}(x^2 - 1) \ln(x^2 - 1) - \frac{1}{2}(x^2 - 1) + C$

✓ **Solution 5: (j)** $\int \frac{x + 2}{(x^2 + 4x + 9)^3} dx$

The numerator is almost the derivative of the quadratic expression inside the parentheses on the bottom. We can use a substitution to turn this fraction into a simple power rule problem.

Step 1: Choose a substitution

We let u be the quadratic expression inside the parentheses:

$$u = x^2 + 4x + 9$$

Now we find the derivative to replace dx :

$$du = (2x + 4) dx = 2(x + 2) dx$$

To match our numerator, we divide both sides by 2:

$$\frac{1}{2} du = (x + 2) dx$$

Step 2: Substitute and integrate

We replace the terms in the integral with u and du :

$$\int \frac{1}{u^3} \cdot \frac{1}{2} du = \frac{1}{2} \int u^{-3} du$$

Using the power rule (add 1 to the exponent and divide by the new number):

$$\frac{1}{2} \left(\frac{u^{-2}}{-2} \right) + C = -\frac{1}{4}u^{-2} + C = \frac{-1}{4u^2} + C$$

Step 3: Back-substitute

Finally, we replace u with our original quadratic expression:

$$\frac{-1}{4(x^2 + 4x + 9)^2} + C$$

Answer: $\frac{-1}{4(x^2 + 4x + 9)^2} + C$

Question 6

The second derivative of a function F has the equation $F''(x) = \ln(x)$. Moreover we know that $F(1) = 1$ and $F'(1) = 2$. Find F .

Solution 6

To find the original function F , we need to integrate the second derivative twice. Each integration will produce a constant of integration, which we will solve for using the given initial conditions.

Step 1: Integrate $F''(x)$ to find $F'(x)$

We integrate the second derivative to get the first derivative:

$$F'(x) = \int \ln(x) dx$$

Using Integration by Parts for $\ln(x)$:

$$f(x) = \ln(x) \Rightarrow f'(x) = \frac{1}{x}$$

$$g'(x) = 1 \Rightarrow g(x) = x$$

Resulting in:

$$F'(x) = x \ln(x) - \int x \cdot \frac{1}{x} dx = x \ln(x) - x + C_1$$

Step 2: Solve for C_1 using $F'(1) = 2$

Substitute $x = 1$ into the equation for $F'(x)$:

$$2 = (1) \ln(1) - 1 + C_1$$

Since $\ln(1) = 0$:

$$2 = -1 + C_1 \Rightarrow C_1 = 3$$

So, $F'(x) = x \ln(x) - x + 3$.

Step 3: Integrate $F'(x)$ to find $F(x)$

Now we integrate the first derivative:

$$F(x) = \int (x \ln(x) - x + 3) dx = \int x \ln(x) dx - \int x dx + \int 3 dx$$

Using Integration by Parts for $\int x \ln(x) dx$:

$$f(x) = \ln(x) \Rightarrow f'(x) = \frac{1}{x}$$

$$g'(x) = x \Rightarrow g(x) = \frac{1}{2}x^2$$

This gives:

$$\frac{1}{2}x^2 \ln(x) - \int \frac{1}{2}x dx = \frac{1}{2}x^2 \ln(x) - \frac{1}{4}x^2$$

Combining all parts:

$$F(x) = \left(\frac{1}{2}x^2 \ln(x) - \frac{1}{4}x^2 \right) - \frac{1}{2}x^2 + 3x + C_2$$

$$F(x) = \frac{1}{2}x^2 \ln(x) - \frac{3}{4}x^2 + 3x + C_2$$

Step 4: Solve for C_2 using $F(1) = 1$

Substitute $x = 1$ into the equation for $F(x)$:

$$1 = \frac{1}{2}(1)^2 \ln(1) - \frac{3}{4}(1)^2 + 3(1) + C_2$$

$$1 = 0 - \frac{3}{4} + 3 + C_2 \Rightarrow 1 = \frac{9}{4} + C_2 \Rightarrow C_2 = -\frac{5}{4}$$

Answer: $F(x) = \frac{1}{2}x^2 \ln(x) - \frac{3}{4}x^2 + 3x - \frac{5}{4}$

? Question 7

We know that a function F satisfies $F'(x) = \frac{e^x}{e^{2x} - 1}$ and that $F(1) = 0$. Find F .

(Hint: use the substitution $u = e^x$!)

✓ Solution 7

We are given the derivative of a function and one specific point. To find the original function, we must integrate the derivative and then use the given point to solve for the constant of integration.

Step 1: Choose a substitution

Following the hint, we let u be the exponential term:

$$u = e^x$$

Now we find the derivative to replace dx :

$$du = e^x dx$$

Rewriting the integral in terms of u , noting that $e^{2x} = (e^x)^2 = u^2$:

$$F(x) = \int \frac{e^x}{e^{2x} - 1} dx = \int \frac{1}{u^2 - 1} du$$

Step 2: Factor the denominator

The denominator is a difference of squares:

$$u^2 - 1 = (u - 1)(u + 1)$$

Step 3: Write the partial fractions with unknown constants

We split the fraction into two simpler parts:

$$\frac{1}{(u - 1)(u + 1)} = \frac{A}{u - 1} + \frac{B}{u + 1}$$

Step 4: Sum the partial fractions

Combine them over a common denominator:

$$\frac{A(u+1) + B(u-1)}{(u-1)(u+1)} = \frac{(A+B)u + (A-B)}{u^2 - 1}$$

Step 5: Equal the numerators

We compare this to our numerator of 1:

$$\begin{cases} A + B = 0 & \Rightarrow B = -A \\ A - B = 1 \end{cases}$$

Step 6: Solve for the unknowns

Substituting $B = -A$ into the second equation gives $2A = 1$, so $A = 1/2$ and $B = -1/2$.

The integral becomes:

$$\int \left(\frac{1/2}{u-1} - \frac{1/2}{u+1} \right) du = \frac{1}{2} \ln |u-1| - \frac{1}{2} \ln |u+1| + C$$

Using log rules and replacing u with e^x :

$$F(x) = \frac{1}{2} \ln \left| \frac{e^x - 1}{e^x + 1} \right| + C$$

Step 7: Solve for C using $F(1) = 0$

Substitute $x = 1$ into our equation:

$$0 = \frac{1}{2} \ln \left| \frac{e^1 - 1}{e^1 + 1} \right| + C \Rightarrow C = -\frac{1}{2} \ln \left(\frac{e-1}{e+1} \right)$$

Answer: $F(x) = \frac{1}{2} \ln \left| \frac{e^x - 1}{e^x + 1} \right| - \frac{1}{2} \ln \left(\frac{e-1}{e+1} \right)$

? Question 8

A company buys fresh shellfish, freezes it and sells it in frozen form. The day price of lobster depends on the number of lobsters caught that day. When q lobsters are caught and sold, the marginal revenue is given by $MR(q) = 8 - 0.0018q^2$ where $0 \leq q \leq 60$.

- (a) Determine the revenue when q lobsters are caught and sold.
- (b) Determine the day price of lobster on a day when 50 are caught and sold.

✓ Solution 8: (a) Determine the revenue function $R(q)$

To find the Total Revenue $R(q)$, we integrate the Marginal Revenue $MR(q)$ given in the problem.

Step 1: Integrate the marginal revenue

We use the power rule for integration:

$$R(q) = \int (8 - 0.0018q^2) dq$$

$$R(q) = 8q - \frac{0.0018q^3}{3} + C$$

$$R(q) = 8q - 0.0006q^3 + C$$

Step 2: Solve for C

Since revenue must be zero if no lobsters are sold ($q = 0$), we substitute these values:

$$0 = 8(0) - 0.0006(0)^3 + C \Rightarrow C = 0$$

Answer: $R(q) = 8q - 0.0006q^3$

✓ Solution 8: (b) Calculate the day price for 50 lobsters

The price p (also known as the inverse demand) is the Total Revenue divided by the quantity q .

Step 1: Find the price formula

Using the revenue function from part (a):

$$p = \frac{R(q)}{q} = \frac{8q - 0.0006q^3}{q}$$

Dividing each term by q simplifies the expression:

$$p = 8 - 0.0006q^2$$

Step 2: Calculate the price for 50 lobsters

Substitute $q = 50$ into our price formula:

$$p = 8 - 0.0006(50^2)$$

$$p = 8 - 0.0006(2500)$$

$$p = 8 - 1.5 = 6.5$$

Answer: 6.5

Question 9

A manager has to decide how many DVD's of a top movie should be ordered. The marginal revenue for the sale of q DVD's is given by $MR(q) = \sqrt{400 - 20q}$ for $5 \leq q \leq 20$. For most movies, 15 DVD's are ordered. What is the additional revenue to be expected when 20 DVD's for this specific top movie are being ordered and sold?

Solution 9

To find the additional revenue, we first determine the general revenue function $R(q)$ by integrating the marginal revenue $MR(q)$. Then, we calculate the difference between the revenue at $q = 20$ and $q = 15$.

Step 1: Integrate to find the revenue function $R(q)$

We set up the indefinite integral of the marginal revenue:

$$R(q) = \int \sqrt{400 - 20q} \, dq = \int (400 - 20q)^{1/2} \, dq$$

Using u -substitution where $u = 400 - 20q$ and $du = -20 \, dq$ (or $\frac{du}{-20} = dq$):

$$R(q) = -\frac{1}{20} \int u^{1/2} \, du = -\frac{1}{20} \left(\frac{u^{3/2}}{3/2} \right) + C$$

$$R(q) = -\frac{1}{20} \cdot \frac{2}{3} (400 - 20q)^{3/2} + C = -\frac{1}{30} (400 - 20q)^{3/2} + C$$

Step 2: Calculate revenue at $q = 20$

$$R(20) = -\frac{1}{30} (400 - 20(20))^{3/2} + C$$

$$R(20) = -\frac{1}{30} (400 - 400)^{3/2} + C = 0 + C = C$$

Step 3: Calculate revenue at $q = 15$

$$R(15) = -\frac{1}{30}(400 - 20(15))^{3/2} + C$$

$$R(15) = -\frac{1}{30}(400 - 300)^{3/2} + C = -\frac{1}{30}(100)^{3/2} + C$$

Since $100^{3/2} = (\sqrt{100})^3 = 10^3 = 1000$:

$$R(15) = -\frac{1000}{30} + C = -\frac{100}{3} + C$$

Step 4: Find the additional revenue

The additional revenue is the difference $R(20) - R(15)$:

$$\Delta R = (C) - \left(-\frac{100}{3} + C\right) = C + \frac{100}{3} - C = \frac{100}{3}$$

Answer: $\frac{100}{3}$

Question 10

The marginal propensity to save is given by $MPS = \frac{dS}{dY} = 0.3 - \frac{0.1}{\sqrt{Y}}$ where Y denotes the income and S the savings. Moreover, nothing is saved when the income equals 81 monetary units. Find an expression for the savings.

Tip: Marginal Propensity to Save

The Marginal Propensity to Save (MPS) is the derivative of the savings function with respect to income (Y).

- Savings (S) = $\int MPS \, dY$
- The constant of integration (C) is determined using a known point (e.g., $S = 0$ at a specific income level).

Solution 10

To find the expression for savings, we integrate the marginal propensity to save and then use the given condition to solve for the constant.

Step 1: Integrate to find the general savings function

We integrate the MPS function with respect to income Y :

$$S = \int \left(0.3 - \frac{0.1}{\sqrt{Y}} \right) dY$$

Rewriting the square root as a power:

$$S = \int (0.3 - 0.1Y^{-1/2}) dY$$

Using the power rule for integration:

$$S = 0.3Y - 0.1 \left(\frac{Y^{1/2}}{1/2} \right) + C$$

$$S = 0.3Y - 0.2\sqrt{Y} + C$$

Step 2: Solve for C using the condition $S(81) = 0$

We are told nothing is saved ($S = 0$) when income equals 81:

$$0 = 0.3(81) - 0.2\sqrt{81} + C$$

$$0 = 24.3 - 0.2(9) + C$$

$$0 = 24.3 - 1.8 + C$$

$$0 = 22.5 + C \Rightarrow C = -22.5$$

Step 3: Write the final expression

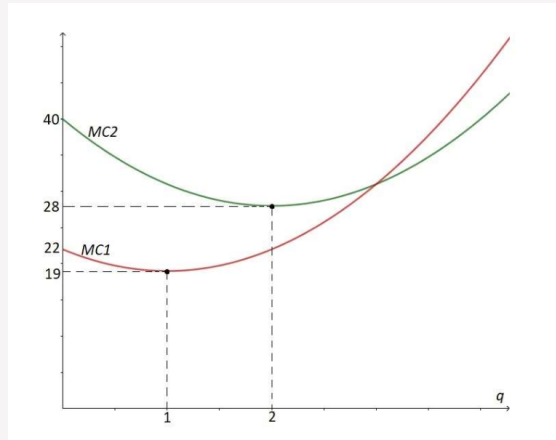
Substituting C back into the function:

$$S = 0.3Y - 0.2\sqrt{Y} - 22.5$$

Answer: $S = 0.3Y - 0.2\sqrt{Y} - 22.5$

Question 11

A company can choose two types of machines to produce a certain good. For the first type, the fixed costs amount to 110 monetary units; for the second type, the fixed costs amount to 230 monetary units. The marginal costs are given by the parabolas below. (The vertices and the intersection points with the vertical axes are given on the figure.)



For which production amounts is the first type of machine the cheapest?

Tip: Finding the Equation from a Graph

If you know the **vertical start** (where the curve hits the y -axis) and the **bottom point** (the vertex), you can find the formula $MC = aq^2 + bq + c$ without complex algebra.

- **Step 1:** The value of c is simply the number where the graph hits the vertical axis ($q = 0$).
- **Step 2:** Find the "steepness" (a) using the bottom point ($q_{\text{bottom}}, y_{\text{bottom}}$):

$$a = \frac{c - y_{\text{bottom}}}{(q_{\text{bottom}})^2}$$

- **Step 3:** Find the middle number (b) using the a you just found and the bottom point's q value:

$$b = -2 \cdot a \cdot q_{\text{bottom}}$$

✓ Solution 11: Determine the production range where the first machine is cheapest

To find the cheapest machine, we first build the formulas for the Marginal Cost (MC) curves using the points shown on the graph.

Step 1: Find the formula for MC_1

Vertical start $c = 22$. Bottom point at $q = 1$ and $y = 19$.

$$a = \frac{22 - 19}{1^2} = 3$$

$$b = -2(3)(1) = -6$$

So, $MC_1 = 3q^2 - 6q + 22$.

Step 2: Find the formula for MC_2

Vertical start $c = 40$. Bottom point at $q = 2$ and $y = 28$.

$$a = \frac{40 - 28}{2^2} = \frac{12}{4} = 3$$

$$b = -2(3)(2) = -12$$

So, $MC_2 = 3q^2 - 12q + 40$.

Step 3: Build the Total Cost functions (TC)

We integrate the MC formulas and add the Fixed Costs ($FC_1 = 110$, $FC_2 = 230$):

$$TC_1 = \int (3q^2 - 6q + 22) dq + 110 = q^3 - 3q^2 + 22q + 110$$

$$TC_2 = \int (3q^2 - 12q + 40) dq + 230 = q^3 - 6q^2 + 40q + 230$$

Step 4: Compare the machines

The first machine is cheaper when $TC_1 < TC_2$:

$$q^3 - 3q^2 + 22q + 110 < q^3 - 6q^2 + 40q + 230$$

$$3q^2 - 18q - 120 < 0 \Rightarrow q^2 - 6q - 40 < 0$$

Factoring $(q - 10)(q + 4) < 0$. Since production q must be positive, this is true for $q < 10$.

Answer: If less than 10 units of the good are being produced.

Chapter 2 - Differential equations

Booklet 2 - Differential equations

Chapter 3 - Definite Integrals

Booklet 3 - Definite Integrals

Chapter 4 - Linear Algebra

Booklet 4 - Linear Algebra

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